Chapter 1: Introduction

1.1 Project Background

Financial literacy helps people make smart choices about saving, spending, budgeting and setting goals. It proves especially vital for young adults who are starting to live on their own. Without it, they could save poorly, get into more debts and make long-term plans that harm economic stability. Mauritius youth deal with this amid scarce formal finance training in schools. A 2024 study found the national financial literacy rate at just 39%, compared to 71% in leading nations like Denmark (Sunechor and Luchoo 2024). Youth joblessness hovered at 23.5 percent back in 2019, a figure that highlights skill shortages addressed by efforts such as Junior Achievement Mascareignes (Rogers Capital 2024).

Financial technology (fintech) progress opens doors for mobile apps to tackle these issues head on. Through gamification, goal tools and simulations, such apps turn dry concepts into hands-on practice. This project builds a targeted app for youth, relying on mock data to cover budgeting, saving and goals without linking to actual bank accounts.

1.2 Problem Statement

Young people in Mauritius frequently feel unsure about handling their money. Standard approaches like school lessons or handouts come across as dull or too abstract to use daily. Research points to weak spots in basics such as saving, where scores average around 52 percent, and borrowing (Sunechor and Luchoo 2024). Apps like Mint or YNAB serve working adults with steady pay and accounts. They skip over students or beginners lacking that foundation. The result leaves a clear opening for an app that mixes safe simulations, lessons and fun elements to build real skills from scratch.

1.3 Aim and Objectives

Aim

The primary aim of this study is to design and develop a fintech mobile application that enhances financial literacy among youth through interactive learning, goal setting, and simulated financial scenarios.

Objectives

The objectives of this study are:

- To analyze the financial literacy challenges faced by youth.
- To review existing fintech and financial education applications and identify their limitations.
- To design a user-centered mobile application focused on financial learning.

- To incorporate features such as onboarding, goal creation, progress tracking, and gamified learning elements.
- To evaluate how the application design supports user engagement and understanding of basic financial concepts.

1.4 Project Scope

The scope of this study is limited to the design and development of a prototype fintech mobile application aimed at improving financial literacy among youth. The application operates using simulated financial data and does not integrate with real banking systems.

The project focuses primarily on:

- Front-end design and user experience.
- Interactive educational modules.
- Virtual budgeting and goal-setting tools.
- Gamified learning mechanisms to enhance engagement.

The study does not include:

- Integration with live financial institutions.
- Real-money transactions.
- Regulatory compliance for financial services.

1.5 Gantt Chart

Chapter 2: Background Study

2.1 Literature review

2.1.1 Financial Literacy: Definition and Concept

Financial literacy is widely recognized as the combination of knowledge, skills, attitudes, and behaviors that enable individuals to make informed and effective decisions regarding their financial resources (Gumilar, Sangka and Totalia, 2024). It includes understanding core financial concepts such as budgeting, saving, credit management, investing, and financial planning, and the ability to apply these concepts in real-world scenarios to achieve financial well-being (Lusardi & Messy, 2023). Importantly, financial literacy is multidimensional, encompassing not only knowledge acquisition but also cognitive skills, behavioral habits, and attitudinal dispositions that collectively empower individuals to manage their finances responsibly.

Financial literacy is not innate or static; rather, it develops progressively over time through formal education, informal learning from family, peers, social networks, and practical experiences in managing money (Rehman & Mia, 2024). Experiential learning plays a key role in reinforcing financial concepts, enabling individuals to internalize knowledge and translate it into responsible financial behaviors. For instance, tracking personal expenses, participating in financial simulations, or engaging with digital financial tools allows learners to practice decision-making in safe environments, enhancing both confidence and competence. A financially literate individual is therefore characterized not only by understanding financial concepts but also by the ability to apply them in everyday life, make informed decisions, set realistic financial goals, and evaluate risks and opportunities effectively (Ahmad and Mohamed Zabri, 2023).

With the rapid digitization of financial services, digital financial literacy (DFL) has become an essential subset of overall financial literacy. DFL encompasses the skills required to access, evaluate, and effectively use digital financial tools, including mobile banking applications, e-wallets, online payment platforms, and other fintech solutions, while being aware of potential risks such as fraud, overspending, and breaches of privacy (Low et al., 2023). The ability to use digital financial tools safely and effectively requires a combination of technical proficiency, behavioral awareness, and critical judgment. Digital financial literacy, in particular, depends not only on access to technology but also on the confidence and skills to navigate digital platforms safely. Integrating digital tools with financial education ensures that learners not only acquire conceptual knowledge but also develop practical skills and responsible behaviors (Gumilar, Sangka & Alfarisy, 2024).

The significance of financial literacy is particularly pronounced among youth and young adults aged 16–30, who are often in the transitional phase toward financial independence. This group faces unique challenges, including limited prior experience with financial products, exposure to aggressive digital marketing, and the temptation of immediate gratification. Developing financial literacy at this stage is critical for fostering confidence, autonomy, and long-term financial resilience. Studies indicate that youth with higher financial literacy exhibit better decision-making, improved budgeting and saving habits, and greater preparedness for future financial challenges (OECD, 2024).

Research also highlights that financial literacy can be strengthened through structured interventions, including educational programs, mobile-based platforms, and gamified learning experiences. Mobile applications and gamified platforms provide interactive, personalized, and engaging opportunities to practice financial decision-making, monitor spending, track savings goals, and receive real-time feedback on choices (Rasyidah, Nawang and Shukor, 2023). Experiential exercises, simulations, and parental involvement further reinforce knowledge, while promoting behavioral change and improving self-efficacy in financial decision-making (Senduk et al., 2024). The integration of digital tools with practical learning exercises allows learners to bridge the gap between theory and application, increasing the likelihood of sustained financial competence.

Financial literacy is influenced by multiple factors, including socio-economic background, education level, generational characteristics, and access to digital technology (Wahyuni, Mauliza & Yanti, 2024). For example, youth from higher-income households may have greater exposure to financial concepts and resources, whereas those with limited access to financial tools require more targeted educational interventions. Parental guidance, peer influence, and social networks also play important roles in shaping financial habits. Integrating financial education with technology-mediated experiences ensures that learners develop both conceptual understanding and practical skills necessary to apply financial knowledge effectively (Rehman & Mia, 2024; Gumilar, Sangka & Alfarisy, 2024).

In conclusion, financial literacy is a multidimensional competency encompassing cognitive knowledge, behavioral skills, and attitudinal awareness, which collectively enable individuals to make informed and effective financial decisions. For youth and young adults, developing both traditional and digital financial literacy is crucial for achieving financial independence, informed decision-making, and long-term resilience. Evidence suggests that mobile-based interventions, gamified learning, experiential exercises, and parental involvement provide practical, engaging, and context-specific opportunities for youth to translate knowledge into responsible financial behavior (Gumilar, Sangka and Alfarisy, 2024).

2.1.2 Mobile App-Based Interventions for Financial Literacy

Mobile applications are increasingly recognized as effective tools for promoting financial literacy and improving financial management skills among youth and young adults. Frisancho et al. (2023) evaluated a budgeting app in Peru, which included SMS reminders and regular enumerator visits over 27 weeks. The study found significant improvements in financial knowledge, including understanding market prices, particularly among youth without prior financial education. While the intervention did not directly increase savings behavior, it encouraged greater use of formal credit and diversified borrowing strategies. These findings demonstrate that mobile-app-based interventions can successfully raise financial awareness and motivate participants to seek additional financial knowledge, even in the absence of formal classroom education.

Similarly, Vangjel et al. (2024) investigated the use of cash management applications among young adults and reported a positive, though modest, relationship between app usage and improved financial outcomes. Users who engaged with these apps were better able to track expenses and control unnecessary spending, highlighting the potential of digital tools to foster budgeting skills. However, the study noted that adoption was constrained by low awareness and limited familiarity with these applications, suggesting that accessibility and promotion are critical for sustained impact.

Targeted personal finance apps also demonstrate strong potential for improving financial literacy. Imawan et al. (2025) developed an Android-based application for university students that provided features such as income tracking, expense monitoring, budgeting, and goal setting. Usability testing with 50 participants revealed a high satisfaction score of 4.6/5, with users praising the app’s intuitive design, accurate tracking, and motivational notifications. The

study emphasizes that apps designed for specific user groups, such as young adults in higher education, can address unique financial challenges and encourage responsible financial behaviors. Feedback from the study suggested that future enhancements, including personalized recommendations and deeper analytical features, could further support informed financial decision-making.

Expanding beyond individual interventions, broader research underscores the importance of financial literacy in promoting effective use of digital financial tools. The European Investment Fund (2024) and Franziska Koefer et al. (2024) investigated how inclusive finance organizations and microfinance institutions (MFIs) across Europe address financial and digital literacy gaps among vulnerable populations. The study identified three core strategies: delivering personalized financial education, providing clear and accessible information, and offering ongoing human support through loan officers and customer service teams. Qualitative analysis indicated that targeted, “just-in-time” financial education through onboarding, workshops, and coaching effectively strengthened participants’ understanding of financial products and services. Moreover, combining digital tools with human guidance helped build confidence, trust, and long-term engagement. The study concluded that technology alone is insufficient; financial literacy initiatives must be complemented by education and support to ensure meaningful adoption and improved financial behaviors.

Similarly, the Advancing Digital Financial Literacy: Insights from two pilot projects in Africa (2025) highlighted the critical role of digital financial literacy (DFL) in low- and middle-income countries (LMICs), particularly in the context of mobile money adoption. Drawing on data from the World Bank’s Global Findex database, regional surveys, and pilot programs in Senegal and Kenya, the report emphasized that lack of user knowledge, confidence, and trust remains a key barrier to effective financial participation. The pilots focused on four design areas: selecting appropriate platforms and channels, delivering localized and easy-to-understand content, building partnerships with trusted institutions, and implementing robust monitoring and evaluation. Results showed that integrating digital literacy training with financial education enabled participants to navigate mobile money systems confidently, reduce financial risk, and engage more actively with formal financial services. These findings underscore the broader impact of mobile-based interventions on financial inclusion, economic empowerment, and gender equity.

Studies targeting students and young adults further reinforce the role of mobile interventions in financial literacy. Silva et al. (2024) conducted a study with 372 college students in the Philippines to examine the relationship between digital financial literacy and responsible use of platforms such as GCash, PayMaya, and PayPal. Using multi-stage sampling and correlational analysis, researchers found that higher literacy levels correlated with improved budgeting, consistent saving, and better financial planning. Despite these gains, students still displayed limited understanding of online savings accounts, fraud prevention, and digital security practices. Recommendations emphasized integrating digital financial literacy into curricula, alongside government-led awareness campaigns, to foster safe and informed digital financial behaviors.

In Oman, a mobile application aimed at enhancing college students' financial literacy combined financial tracking, machine learning predictions of income and expenditure, and contextual recommendations (Kamarudeen and Vijayalakshmi, 2023). The study employed Android Studio, Flutter, and Firebase to develop an app tailored to student needs, incorporating agile development principles and user feedback. The app's predictive analytics and visualization features allowed students to forecast financial outcomes, identify spending patterns, and make informed decisions. Evaluation demonstrated that integrating education with practical, context-specific tools increased both financial knowledge and proactive financial behaviors. Limitations included scalability challenges, but the study highlighted the potential for adaptive technologies to bridge literacy gaps effectively.

Evidence from India and Vietnam confirms the link between digital financial literacy and improved financial behaviors. In India, Chhillar, Sharma and Arora (2025) conducted a study using Partial Least Squares Structural Equation Modeling (PLS-SEM) on survey data from 499 participants, revealing that digital financial literacy strongly influenced personal financial management behaviors and overall financial well-being. The research highlighted that confidence in future financial planning had a greater psychological impact than current financial stress, emphasizing the role of literacy in resilience and stability. Similarly, Ha, Sensoy and Phung (2023) explored the interplay between financial literacy, perceived trust, and mobile money adoption in Vietnam. Survey findings showed that while financial literacy alone supported adoption, trust in the technology and awareness of risks were equally critical. These studies demonstrate that mobile-based interventions must address both knowledge and behavioral confidence to achieve meaningful improvements in financial literacy.

Collectively, these studies suggest that while mobile apps improve awareness and financial discipline, their impact depends on usability, accessibility, and sustained adoption. Evidence shows that mobile app-based interventions enhance financial knowledge, foster budgeting and goal-setting skills, and encourage responsible financial behaviors. The most effective programs combine user-friendly digital tools with personalized education, guidance, and ongoing support, while tailoring content to local contexts and the specific needs of different populations. Although challenges remain particularly in populations with limited prior exposure to digital or financial services, mobile applications offer a promising pathway for promoting long-term financial competence, inclusion, and empowerment.

2.1.3 Game-Based and Simulation Approaches

Game-based and simulation-based approaches have emerged as innovative strategies to enhance financial literacy by engaging learners through interactive, experiential methods. Rather than relying solely on traditional lectures or rote learning, these approaches immerse participants in realistic financial decision-making scenarios, promoting deeper understanding, motivation, and behavioral change.

Yi et al. (2025) explored game-based learning among Malaysian primary school students, demonstrating how interactive activities such as distinguishing between needs and wants, comparing prices, and playing a modified version of the Cash Flow game significantly improved children's financial understanding. The study found that more than 95% of learners developed a clear grasp of value concepts, highlighting that playful learning at an early age can effectively build foundational financial skills and responsible money habits. Similarly, Estima and Peguinho (2024) implemented the *ISCA Family Game* among Portuguese secondary school students, emphasizing the value of role-playing and household decision-making. Their results showed that gamified learning fosters sustainable consumer behavior and better financial planning by enabling students to practice real-world decisions in a safe, engaging environment.

In higher education, simulation-based financial education has gained traction as an effective experiential tool. Suryadi, Sumetri and Prayustika (2024) conducted a study at Bali State Polytechnic that integrated digital simulation games into business and accounting courses, where students managed virtual companies, budgeted expenses, and analyzed profits. Findings revealed a substantial increase in financial management competence and motivation, illustrating how simulations help bridge the gap between theoretical learning and practical financial application. Likewise, Wahid Hasyim Kosasih and Agus Juhana (2025) examined the use of investment simulation platforms at Undiksha University to train students in stock trading and portfolio management. The results demonstrated marked improvements in risk analysis and investment literacy, reinforcing that simulations cultivate applied financial skills and decision-making confidence.

Marta et al. (2024) expanded this concept internationally with the *Financial Escape Room*, a digital simulation tested in Portugal, Italy, Spain, and Lithuania involving over 2,000 participants. The randomized trial revealed a 0.313 standard deviation improvement in literacy scores and underscored the scalability of simulation-based interventions in promoting critical thinking and collaborative problem-solving. In parallel, Lisana et al. (2024) examined META-Z, a hybrid financial literacy initiative that merged a board game and mobile app to teach Generation Z, showing strong cognitive and behavioral improvements. The hybrid format proved effective for digital-native learners, combining tactile play with digital convenience to sustain engagement and encourage responsible financial decision-making.

Ramirez-Montoya et al. (2025) further validated the use of immersive environments through Serious Games, where university students managed personal budgets, set savings goals, and navigated financial challenges. Participants not only demonstrated enhanced financial planning skills but also exhibited positive shifts in financial attitudes and self-efficacy. Similarly, Ramesh et al. (2025) examined an Omani study on Gamification in Accounting Education, which confirmed that introducing point systems, leaderboards, and interactive quizzes increased student motivation and conceptual understanding of accounting and financial management principles.

Finally, Frisancho et al. (2023) contributed a technological dimension by showing how mobile-app-based behavioral interventions could improve financial awareness among out-of-school youth in Peru. While not a traditional "game," the app incorporated behavioral

nudges and interactive budgeting features that simulated financial planning, resulting in improved knowledge of credit use and market prices. This highlights the growing intersection between gamified design and mobile learning in expanding access to financial education.

Collectively, these studies affirm that game-based and simulation-based financial education provides an engaging, hands-on approach that enhances both cognitive and behavioral dimensions of financial literacy. By integrating fun, challenge, and realism, these tools foster motivation and retention while preparing learners to apply financial knowledge in everyday life. However, as several researchers note, the long-term impact of such interventions on sustained financial behavior remains a vital area for future research.

2.1.4 Gamification and Engagement Mechanisms

While game-based learning tools often rely on structured gameplay to deliver educational outcomes, a substantial body of research now emphasizes gamification as a critical mechanism for promoting financial literacy and sustained user engagement. Unlike complete game environments, gamification refers to the integration of game elements such as points, rewards, leaderboards, progress tracking, and challenges into non-game contexts to increase user motivation and participation. In the context of financial education, these mechanics serve to make abstract or complex financial concepts more tangible, enjoyable, and behaviorally impactful.

Yulianto, A.Y. et al. (2024) investigated the influence of gamification on financial education among university students, focusing on its effects on enjoyment, engagement, and knowledge acquisition. Their quantitative study, which applied Variational Bayesian Structural Equation Modeling (VB-SEM), revealed that although gamification positively affected both enjoyment and engagement, only engagement mediated the relationship between gamification and financial knowledge. This distinction is significant; it suggests that the motivational aspects of gamification are not, by themselves, sufficient to promote deeper learning unless they lead to active cognitive participation. Therefore, the study supports the argument that gamification's educational impact lies primarily in its ability to sustain meaningful engagement rather than superficial amusement. In practice, this implies that designers of financial literacy programs must focus on interactive features that encourage learners to think critically and apply knowledge, rather than simply rewarding participation.

In a similar vein, Dr. Rajsee (2024) explored the application of gamification in fintech applications targeted at young investors aged 25–34. The study found that gamified features such as cashbacks, prizes, and point systems were highly effective in increasing user engagement and accessibility to financial tools. However, it also raised ethical and behavioral concerns, noting that these features might encourage excessive risk-taking and compulsive engagement. The study therefore presents a nuanced view of gamification while it can drive participation and improve financial awareness, it must be carefully managed to prevent counterproductive financial behaviors. This aligns with Yulianto, A.Y. et al.'s conclusion that engagement must be not only high but also purposeful, reinforcing the importance of ethical and responsible gamification in financial platforms.

The gamification of financial learning is not limited to adult or university learners. Estima and Peguinho (2024) introduced the *ISCA Family Game*, a gamified initiative designed for students aged 12–18 to promote both financial literacy and consumer education. The project used interactive role-play and problem-solving to engage students in topics such as budgeting, saving, and consumption choices. The game's success in attracting educators' and parents' interest underscores gamification's potential in early education, particularly as a tool to foster responsible financial behavior from a young age. The authors argue that such initiatives can act as prototypes for other institutions seeking to incorporate gamified tools into classroom or extracurricular activities, thereby cultivating the next generation of financially literate individuals through experiential learning.

Similarly, Ruiz-Carhuamaca, et al., (2024) designed a mobile learning app that incorporates gamified educational challenges to teach financial concepts to young people. Their approach combined user experience design principles with game mechanics such as levels, progress feedback, and badges to ensure an engaging and accessible learning process. The study found that users not only reported greater enjoyment but also demonstrated a measurable improvement in their understanding and application of financial principles. The authors concluded that gamification could bridge the gap between theoretical financial education and real-life financial behavior, empowering users to make more informed financial decisions. This reinforces the idea that the motivational pull of gamification can serve as a vehicle for deeper cognitive engagement and behavioral transformation.

Ramesh et al. (2025) examined gamification in accounting education and found comparable results. By integrating points, badges, and leaderboards within university accounting courses, the researchers observed marked improvements in student focus, participation, and enthusiasm. Importantly, engagement rather than simple exposure to game features was again identified as the critical mediator linking gamification to improved learning outcomes. The study also highlighted demographic differences, noting that female students and high-performing learners responded more positively to competitive and reward-based systems. These findings suggest that the success of gamification depends on tailoring motivational structures to the preferences and cognitive profiles of different learner groups.

In a complementary study, Supathanarangsri et al. (2020) developed a mobile application for financial literacy that incorporated game mechanics such as badges, levels, and real-time feedback to maintain user motivation. Their results demonstrated that interactive and rewarding gameplay encouraged sustained engagement and repeated use of the app, which led to improved retention of financial knowledge. However, the authors cautioned against over-reliance on extrinsic rewards, emphasizing the importance of fostering intrinsic motivation such as curiosity, autonomy, and self-improvement to ensure long-term learning outcomes. This aligns closely with Yulianto, A.Y. et al. (2024), who emphasized engagement as the bridge between motivation and genuine knowledge acquisition.

Şenol and Onay (2022) extended this discourse to investment education by examining stock market simulation games as a form of gamified learning. Their findings showed that repeated exposure to simulated trading reduced overconfidence bias and improved analytical

decision-making among participants. The use of leaderboards and progress indicators encouraged reflective engagement, transforming the activity from mere play into a structured learning process. However, they also observed that some users developed familiarity biases toward certain assets, illustrating that gamified systems must be carefully calibrated to avoid reinforcing cognitive distortions. This highlights that while gamification is a powerful engagement tool, its psychological effects must be monitored to ensure that the learning environment remains constructive.

Finally, Liu et al., (2024) analyzed the effects of gamification within microfinance platforms, focusing on how it influenced user engagement, satisfaction, and retention. Through Partial Least Squares Structural Equation Modeling (PLS-SEM), they found that reward points, milestones, and achievement badges significantly enhanced user loyalty and sustained participation. However, when game elements became overly complex or burdensome, user engagement declined. This finding underscores the importance of simplicity, transparency, and goal alignment in the design of gamified financial systems to maintain their motivational impact.

Collectively, these studies illustrate that gamification serves a dual function in financial literacy initiatives: it is both a motivational catalyst and a behavioral regulator. When well-designed, gamification fosters curiosity, persistence, and enjoyment that translate into deeper cognitive processing and improved financial behavior. However, when misapplied through excessive rewards or unbalanced competition, it risks undermining the educational objectives it seeks to achieve. Therefore, the most effective use of gamification in financial literacy lies in achieving balance: combining extrinsic motivators like rewards and progress tracking with intrinsic motivators such as curiosity, autonomy, and mastery. By doing so, financial learning experiences become not only engaging but also transformative encouraging sustained participation and long-term financial competence.

2.1.5 Design Thinking and User-Centered Approaches

Beyond content and gamification, the design of financial literacy applications strongly influences their success. While engaging visuals and interactive features may capture initial attention, it is the underlying design philosophy, particularly Design Thinking and user-centered approaches, that determines whether users stay engaged, absorb financial concepts, and apply them meaningfully. These approaches prioritize empathy, personalization, and iterative development, ensuring that applications are tailored to users' needs, minimize cognitive load, and foster sustained motivation.

Zhang (2021) exemplifies the application of Design Thinking in financial literacy through the creation of a digital financial services (DFS) app. The development process followed the five phases: empathize, define, ideate, prototype, and test. It also incorporated behavioral nudges such as facilitator, spark, signal, and inquiry. These nudges were designed to guide users through complex content while maintaining engagement, demonstrating how thoughtful design can enhance learning outcomes.

Ruiz-Carhuamaca et al. (2024) proposed a gamified framework that addressed youth financial literacy gaps by emphasizing personalization, interactive challenges, and user experience. Their model aligns with Design Thinking principles by identifying user pain points and iteratively designing solutions that foster motivation and retention. Similarly, Nguyen (2022) highlighted the importance of user experience in financial literacy apps, arguing that intuitive interfaces and personalized feedback loops are essential for deep engagement and long-term learning.

Empirical studies reinforce these conceptual models. Imawan et al. (2025) developed a finance management app for university students using the waterfall development model. The app featured budgeting tools, goal tracking, and motivational prompts, and was rated highly for usability and effectiveness. Students reported increased financial awareness and a greater sense of control over their spending habits.

Muthusamy et al. (2023) introduced FinEduPlay, a gamified savings app for children that included a virtual piggy bank, quiz leveling system, and customizable avatars. Their findings showed unanimous agreement among users on the app's ease of use and motivational impact. The design emphasized simplicity, visual appeal, and reward mechanisms to encourage saving behavior among young users.

Aminuddin, Zain and Aziz (2024) applied the Mechanics-Dynamics-Aesthetics (MDA) framework and Agile methodology to create MoneyMaster, a gamified financial literacy app for young adults. The app achieved an 84.4% enjoyment score on the EGameFlow scale, validating its engaging and educational design.

Pillai et al. (2025) explored gamified learning as a tool for enhancing financial literacy across diverse age groups. Their study emphasized the importance of tailoring game mechanics to different user segments including children, youth, and adults, by adjusting complexity, feedback styles, and motivational triggers. The app design incorporated real-time feedback, progress tracking, and adaptive challenges, showing that gamification can be both educational and inclusive when grounded in user-centered design principles.

Deshmukh, Madachane, and Pawar (2025) developed FinAI, a gamified finance learning platform that integrated artificial intelligence to personalize the learning experience. The platform used AI-driven analytics to adapt content difficulty and provide targeted feedback based on user performance. This approach enhanced engagement and retention by aligning financial education with individual learning styles and behavioral patterns.

Frisancho et al. (2023) conducted a large-scale behavioral intervention in Peru, demonstrating that even minimalistic tools such as budget recording features paired with SMS nudges could improve financial literacy among youth with no prior exposure. This supports the idea that user-centered design does not require complexity, but rather strategic alignment with user behavior and context.

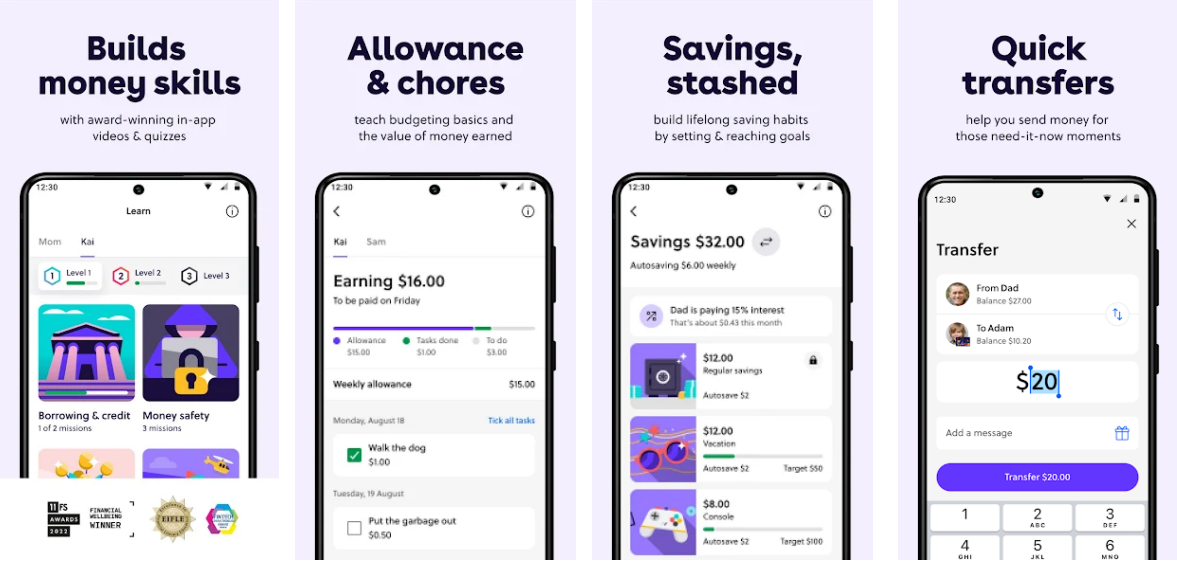
Finally, Ongwae and Duncombe (2021) examined mobile money innovations in Kenya, emphasizing the importance of user involvement in digital systems design. Their case studies

revealed that inclusive design processes, especially those that consider marginalized populations, are critical for adoption and long-term impact. They argue that financial tools must be co-created with users to ensure relevance, trust, and usability.

In conclusion, the collective evidence from these studies underscores that Design Thinking and user-centered approaches are foundational to the success of financial literacy applications. Whether through behavioral nudges, gamified interfaces, experiential learning, or inclusive design processes, these methodologies ensure that financial tools are not only informative but also engaging, accessible, and responsive to diverse user needs. By aligning educational objectives with intuitive and personalized design, developers can create applications that truly empower users to build lasting financial competence.

2.2 Existing Systems

2.2.1 GoHenry by Acorne Kids Banking



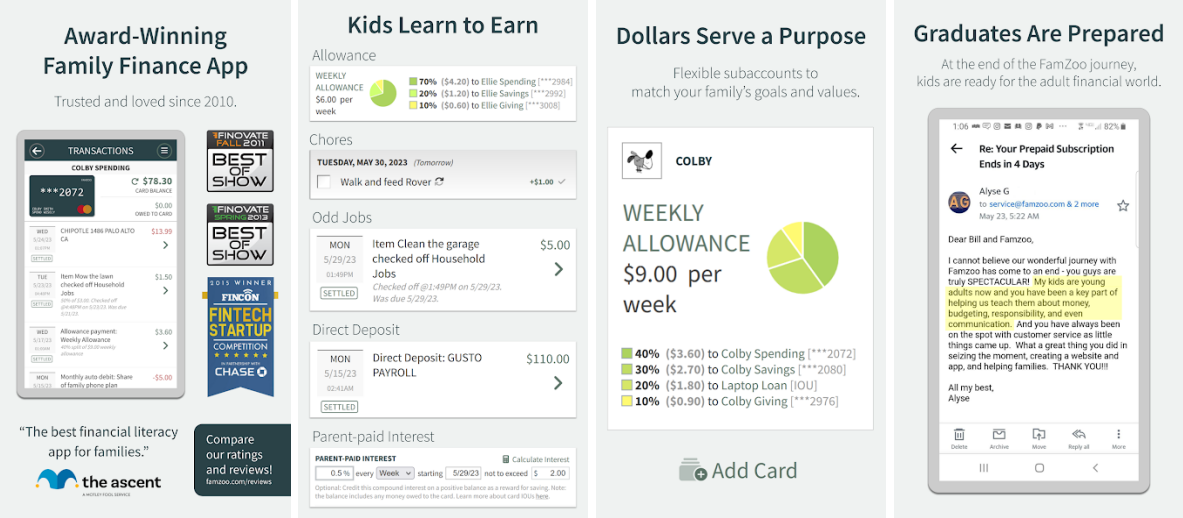
GoHenry by Acorns Kids Banking (Google Play) is the debit card & banking app for kids and teens. Young people aged 6-18 can learn about earning, saving, and smart spending with features like automated allowance, task lists, and gamified educational quizzes and videos. Then they can practice their skills in the real world with their own debit card. All while parents watch and guide with flexible parental controls. That's why 92% of parents say their children are more money-confident since joining GoHenry.

Some interesting features of GoHenry are:

- Automated allowance – Kids receive money automatically for chores or tasks.
- Chore and task lists – Encourages responsibility and earning habits.
- Gamified educational quizzes and interactive videos – Makes learning about money fun and engaging.

- Real-world practice with debit card – Teens can spend and save using their own card.
- Flexible parental controls – Parents can monitor and guide spending.
- Goal-setting and spending insights – Helps users plan savings and track expenses.
- Customizable debit cards – Adds a personal touch and motivates engagement.

2.2.2 FamZoo Family Finance

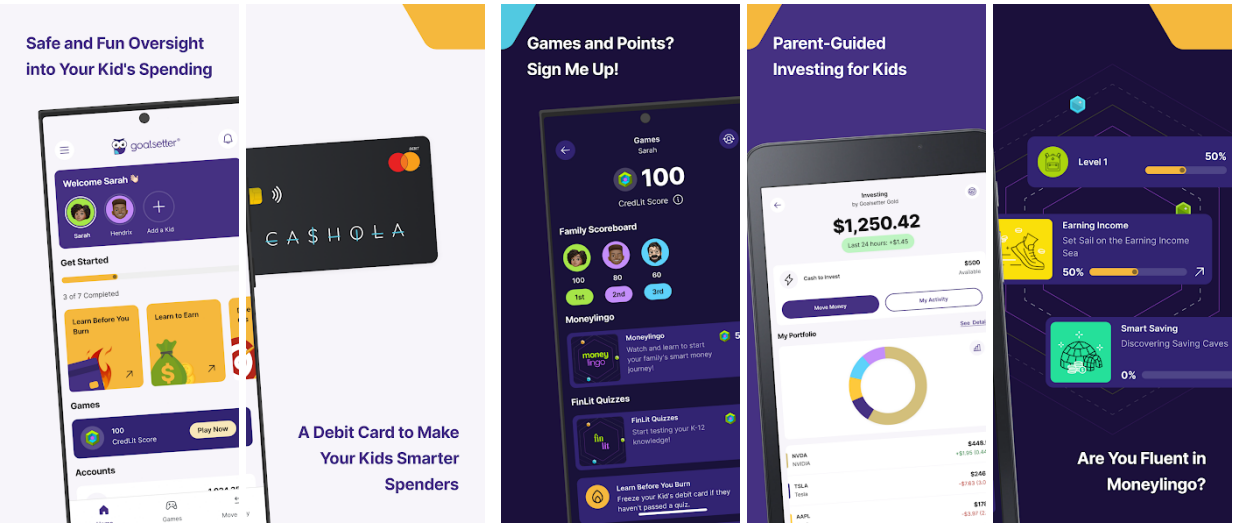


FamZoo (Google Play) helps kids, teens, and parents develop good personal finance habits without the hassle. FamZoo is a collection of prepaid cards for your family. The cards (U.S. only) are linked together by our award-winning app filled with family finance features that reinforce good money habits. (If your kids aren't quite ready for our cards or live outside the U.S., you can use FamZoo's IOU accounts to track money held elsewhere.) Use FamZoo to manage your family's spending, saving, charitable giving, allowances, chores, budget…

Some interesting features of FamZoo are:

- Virtual family bank – Allows parents to create a private banking system for kids.
- Allowance management – Automates allowances with flexible schedules.
- Goal-setting and savings jars – Kids can set savings goals and allocate money into virtual jars.
- Chore tracking – Parents assign chores and reward completion with funds.
- Prepaid cards for kids – Teens can practice spending safely in the real world.
- Parental controls and oversight – Parents monitor transactions, set limits, and guide financial habits.
- Educational focus – Promotes money management skills through practice and structured family banking.

2.2.3 Goalsetter: Invest & Bank



Goalsetter (Google Play) is a mobile banking, debit card that focuses on educating the next generation with fun financial quizzes based on pop culture, putting them on the path to financial freedom. Whether you're a teen wanting financial independence, a parent wanting to start saving with their kids, or an adult who wants to jump-start their financial literacy, we have features for every need.

Some interesting features of Goalsetter are:

- “Learn Before You Burn” quizzes – Kids must complete educational quizzes before spending money.
- Savings goals and virtual wallets – Encourages setting goals and tracking progress.
- Badges and certificates – Rewards learning achievements and motivates continued engagement.
- Parental oversight – Parents can monitor progress and guide financial decisions.
- Interactive lessons and games – Makes financial education engaging through gamification.
- Community challenges – Users can participate in group challenges to earn rewards and learn collaboratively.
- Customizable debit cards – Provides real-world practice while maintaining safety.

2.2.4 FitMoney $upersquad

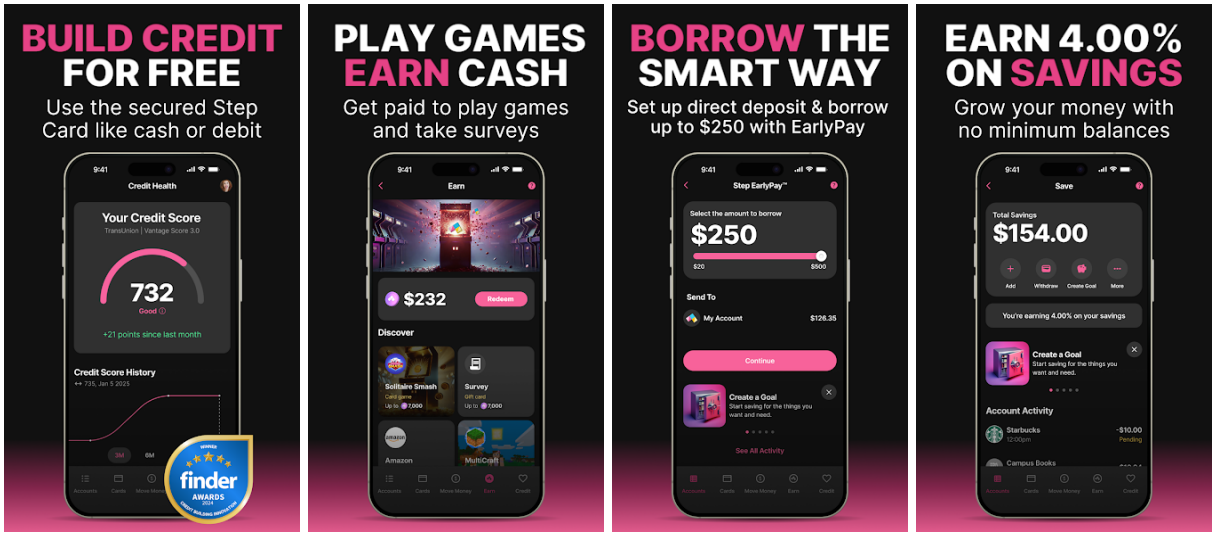


FitMoney $uperSquad is a free web-based interactive financial literacy game designed for children. It teaches the basics of money management through videos, quizzes, and mini-games. Students navigate a virtual world called Fitonia, starting with the Meet Your Money module to learn about money concepts, valuing money, and recognizing coins and bills. Subsequent modules cover saving and spending, credit and banking, budgeting, insurance, earnings and taxes, and investing. Upon completing modules, students earn certificates and learning guides. The platform also allows personalization of avatars using coins earned in gameplay, adding a gamified and engaging experience. (Anon., 2022)

Some interesting features of FitMoney $uperSquad are:

- Interactive modules – Covers topics like saving, spending, budgeting, banking, insurance, earning, taxes, and investing.
- Mini-games and quizzes – Reinforce learning through play and multiple-choice questions.
- Certificates and learning guides – Reward students and provide summaries of topics covered.
- Avatar personalization – Kids can customize their avatars using coins earned in the game.
- Audio support – Text can be read aloud to aid understanding.
- Gamified progression – Encourages engagement and completion of modules.
- Virtual world experience (Fitonia) – Makes financial learning immersive and fun.

2.2.5 Step: The All-In-One Money App



Step(Google Play) is the ultimate all-in-one money app for Gen Z. Build credit for free, borrow money when you need it most with no interest, earn cash playing games & taking surveys, and unlock 4% on your savings—all in one app that is trusted by over 6.5 million users.
Build credit, save money and get ahead starting as early as your first paycheck. Step is financial independence at your fingertips.

Some interesting features of Step are:

- No-fee debit card – Teens can spend money safely without worrying about fees.
- Credit building – Helps users start building a positive credit history from age 13.
- Cashback rewards – Encourages smart spending by giving rewards on purchases.
- High-yield savings – Motivates users to save with competitive interest rates.
- Early pay access – Teens can receive money faster through early deposits.
- Parental controls – Parents can monitor spending and guide financial habits.
- Educational tips – Provides guidance on budgeting, saving, and credit management.

2.3 Tools and Technologies

2.3.1 Mobile Application Development

Mobile application development refers to the process of creating software applications that run on mobile devices such as smartphones and tablets. Over the past decade, mobile applications have become central to the digital economy, supporting diverse functions such as education, communication, entertainment, and financial management (Gartner, 2024). The rise of

smartphones has created a demand for applications that are efficient, user-friendly, and compatible across multiple devices.

With the increasing usage of smartphones, mobile applications have become a vital part of digital transformation. There are mainly two dominant operating systems used for mobile app development: Android and iOS.

- Android Operating System
  Developed by Google, the Android operating system is currently the most widely used mobile platform in the world. As open-source software, Android allows developers to modify, customize, and add new features to enhance functionality and compatibility with the latest technologies. This flexibility has contributed to its popularity among manufacturers and developers. Android supports a wide range of technologies such as GSM, 4G/5G, Wi-Fi, Bluetooth, GPS, and NFC, which make it suitable for diverse applications and devices. Applications are mainly developed using Android Studio, which provides an integrated environment for coding, testing, and debugging.

- Iphone OS (IOS)
  Developed by Apple Inc., iOS is the operating system that powers Apple products such as the iPhone and iPad. It provides a secure, efficient, and stable environment due to its closed-source nature and integration with Apple hardware. The main programming languages used for iOS app development are Swift and Objective-C, supported through Xcode, Apple's official IDE. iOS offers features such as Wi-Fi, Bluetooth, VPN support, and seamless synchronization with Apple's cloud service, iCloud. Its strict quality standards and uniform device ecosystem help ensure high performance and security.

2.3.2 Programming Languages and Frameworks

Android Platforms

- **Java**
  Java remains one of the most widely used programming languages for Android application development due to its stability, security, and scalability. It is an object-oriented language that supports code reusability and flexibility, making it suitable for complex applications. Java also offers platform independence through its Write Once, Run Everywhere (WORA) capability, which allows code to be executed on different devices without modification. Its extensive libraries, mature ecosystem, and large developer community make it a reliable choice for building Android apps.

- **Kotlin**
  Kotlin, introduced by Google in 2017 as the official Android programming language, provides a more modern and concise alternative to Java. It is designed to improve

developer productivity with features like null safety, type inference, and coroutines for asynchronous operations. Kotlin is fully interoperable with Java, allowing developers to integrate it into existing Java projects seamlessly. Its compact syntax reduces boilerplate code, enhancing readability and maintainability. Today, Kotlin is widely adopted for Android app development due to its efficiency, safety, and ease of use.

- **C/C++**
  C and C++ are also used in Android development, primarily for performance-critical components. These languages support object-oriented programming and dynamic memory allocation, enabling faster execution of complex tasks. They are highly portable and suitable for cross-platform development, particularly when integrating low-level features or optimizing computationally intensive processes. C++ is compiler-based, which enhances execution speed and efficiency, making it ideal for high-performance modules in mobile applications.

iOS Platforms

- **Swift**
  Swift, introduced by Apple in 2014, is the official programming language for iOS app development. It features a modern, concise syntax that simplifies coding and reduces the likelihood of runtime errors. Swift is open-source and supported by a large developer community, providing extensive resources for development. Its integration with Apple's frameworks, such as SwiftUI and UIKit, enables developers to design interactive and visually appealing applications efficiently. Swift's automatic memory management and type safety mechanisms improve app reliability and performance.

- **Objective-C**
  Objective-C, developed in the early 1980s, remains in use, particularly for maintaining legacy iOS applications. As an extension of the C language, Objective-C supports dynamic typing and message passing, offering flexibility in complex architectures. Despite being more verbose than Swift, Objective-C provides stable performance and seamless API integration. Many established iOS apps still rely on Objective-C components, making knowledge of this language valuable for long-term project maintenance.

Cross-Platform Frameworks

- **Flutter**
  Flutter, introduced by Google in 2017, is a popular open-source framework for building cross-platform mobile, web, and desktop applications using a single codebase. It uses the Dart programming language and is well-known for its “hot reload” feature, which allows developers to see real-time updates during development without recompiling the entire project. Flutter’s widget-based architecture provides extensive flexibility in designing customized user interfaces, supporting highly interactive and visually rich applications. Its compilation to native machine code ensures performance comparable to native applications, while third-party integrations allow access to platform-specific functionalities. Flutter has gained rapid adoption among developers due to its ease of use, growing community support, and suitability for complex, interactive applications such as fintech and educational apps.

- **React Native**
  React Native, developed by Meta, is an open-source framework that enables cross-platform application development using JavaScript and the React library. React Native employs a “bridge” to connect JavaScript code with native modules, allowing applications to access device functionalities such as cameras, sensors, and GPS. This approach provides near-native performance while enabling code reuse across Android and iOS platforms. React Native is recognized for reducing development time and maintenance costs and offers a large repository of pre-built components, which simplifies UI design. Its widespread adoption by major companies, including Facebook and Uber, demonstrates its reliability and scalability for commercial applications.

- **Xamarin**
  Xamarin, maintained by Microsoft, leverages the C# language and the .NET framework to create applications that run on Android, iOS, and Windows platforms. It allows developers to share a significant portion of their codebase across platforms, reducing duplication and improving development efficiency. Xamarin provides near-native performance and offers integration with Visual Studio for code editing, debugging, and testing. Its Xamarin Test Cloud allows developers to test applications on thousands of virtual and physical devices, generating detailed performance reports. This makes Xamarin particularly suitable for enterprise-level applications that require stability, scalability, and long-term maintenance.

2.3.3 Databases

The database of a mobile application is essential for storing and managing all data required for the application’s performance. For a fintech app aimed at promoting financial literacy among youth, the choice of database technologies is critical to ensure secure storage, efficient retrieval,

real-time synchronization, and offline access. Different databases can be used to meet these diverse requirements, depending on the type of data and functionality needed.

- **SQLite**
  SQLite is a lightweight, embedded relational database management system written in C. It operates locally within the mobile application, which allows users to access app features without an internet connection. This is particularly useful for offline access to interactive lessons, quizzes, and personal progress tracking, ensuring that users can continue learning even when connectivity is limited. SQLite's self-contained architecture reduces dependency on the operating system, improving performance and reliability (Li & Chen, 2022).

- **Firebase**
  Firebase is a cloud-based NoSQL database supported by Google that enables real-time data synchronization across devices. It is suitable for features such as leaderboards, streak tracking, badge sharing, and social interactions, where instant updates are required for all connected users. Firebase also provides integrated authentication, analytics, and push notifications, which simplify backend management and support engagement mechanisms, enhancing the gamification elements of the app.

- **PostgreSQL**
  PostgreSQL is an open-source relational database designed for high reliability, scalability, and complex query support. It is suitable for storing structured financial data, such as user budgets, simulated transactions, and progress in interactive lessons. Its ACID-compliance ensures data integrity, while its advanced indexing and transaction mechanisms allow secure, multi-user access. PostgreSQL is therefore well-suited for managing the core financial functions of the app.

- **MongoDB Atlas**
  MongoDB Atlas is a cloud-based NoSQL database that stores data in flexible, JSON-like documents. It is ideal for handling unstructured or semi-structured data, including gamification logs, community challenges, and achievement badges. The database supports global distribution, automated backups, and high availability, ensuring reliable access for users across different regions. Its flexible schema complements other databases by efficiently managing evolving or dynamic app content.

2.3.4 APIs and Services

Application Programming Interfaces (APIs) allow mobile applications to connect with various external services and tools. They enable developers to integrate advanced functionalities

without building everything from scratch. For a fintech mobile learning application, APIs are essential for user management, analytics, data storage, and engagement features.

- **Firebase API**
  Firebase, developed by Google, is a comprehensive backend platform that supports various mobile and web application services. It provides an efficient solution for real-time data synchronization, user authentication, and cloud data storage. The Firebase Realtime Database ensures that any change in data is instantly reflected across all connected devices, enhancing user experience and performance. Additionally, Firebase offers built-in Crashlytics and Cloud Messaging, which help developers monitor app stability and send notifications to users.

- **Firebase Analytics / Google Analytics API**
  Analytics play a vital role in understanding user interactions and improving application performance. Firebase Analytics, integrated with Google Analytics, enables developers to track user activities, such as lesson completions, daily logins, or time spent on each module. The insights gathered from analytics can be used to enhance the design and usability of the app. It also supports event tracking, allowing for data-driven decision-making to optimize user engagement and retention

- **Cloud Storage API**
  The Cloud Storage API, often used through Firebase Cloud Storage or Google Cloud Storage, provides a scalable and secure solution for storing large media files such as images, videos, or certificates. It supports fast uploads and downloads, ensuring that users can easily access learning materials anytime. Access control rules can also be defined to maintain data privacy and integrity.

- **Social Login and Sharing API**
  To promote accessibility and social engagement, the integration of Google Sign-In or Facebook Login APIs allows users to register or log in quickly using existing social accounts. This simplifies the authentication process and enhances user convenience. Additionally, social sharing features can be implemented using built-in Android or iOS sharing intents, allowing users to share achievements, badges, or certificates on social media, thereby motivating peer participation and healthy competition.

2.3.5 Other Tools and Technologies

The development of a cross-platform fintech mobile application requires the use of a range of tools and technologies that support the various phases of the software development lifecycle, from interface design to coding, testing, and version control. The selection of appropriate tools ensures a smooth, efficient, and well-coordinated development process.

- **Visual Studio Code**
  Visual Studio Code (VS Code) is a lightweight, cross-platform Integrated Development Environment (IDE) developed by Microsoft. It supports a wide range of programming languages and frameworks, including Dart and Flutter, which will be used in this project for cross-platform application development. VS Code provides features such as syntax highlighting, debugging, IntelliSense, and built-in Git integration, making it ideal for both front-end and back-end development. Its vast collection of extensions, including those for Flutter, Firebase, and REST APIs, further enhances productivity. Given its flexibility and compatibility with macOS, it will be the primary IDE used for developing the fintech learning application.

- **Android Studio**
  Android Studio, developed by Google, is the official IDE for Android application development. Although the main development will take place in VS Code, Android Studio will be utilized to manage Android SDKs, configure virtual devices, and perform emulator testing for the Android version of the application. The IDE provides comprehensive tools for debugging, profiling, and performance analysis, ensuring that the Android build of the app runs efficiently across various devices and operating system versions.

- **Xcode**
  Xcode, developed by Apple, is the official IDE for iOS application development. Since the app will also be deployed on iOS devices, Xcode will be required for building, running, and testing the iOS version of the application. It provides simulators for multiple Apple devices, ensuring compatibility across different iPhone and iPad models. While Flutter code will not be written directly in Xcode, the IDE is necessary for compiling iOS builds, managing certificates, and deploying the application to physical devices or the Apple App Store.

- **Figma**
  Figma is a cloud-based collaborative design tool that will be used during the UI/UX design phase of the project. It allows designers and developers to create wireframes, prototypes, and user interface layouts in real time. Its collaborative features enable feedback and iteration, ensuring that the app design aligns with usability and aesthetic principles suitable for the youth target audience. Figma supports responsive layouts and interactive components, making it suitable for both Android and iOS design systems.

- **Git and GitHub**
  Git is a distributed version control system used to track code changes, while GitHub is a cloud-based platform that hosts Git repositories. Both tools will be used to maintain version control, allowing the developer to record changes, revert to previous versions, and collaborate efficiently. Using GitHub ensures code backup, transparency, and coordination, especially beneficial in managing project milestones and avoiding data loss.

- **Postman**
  Postman is a popular tool for API testing and documentation. It allows developers to send requests to APIs and examine the responses to ensure that all backend services, such as Firebase or analytics APIs, are functioning correctly. Postman will be used to test the application's communication with external services and verify data transfer consistency during development.

Chapter 3: Analysis

3.0 Introduction

This chapter presents a detailed system analysis for the proposed financial literacy application. The purpose of this chapter is to analyse existing systems, identify their limitations, and justify the need for the proposed solution. A comparative analysis is conducted to highlight the gaps in current financial education applications. The chapter also defines the functional and non-functional requirements of the system, outlines the system users, and provides an overview of the proposed system. This analysis forms the foundation for the system design presented in the next chapter.

3.1 Analysis of Existing Systems

In recent years, several mobile applications have been developed to promote financial awareness among children and young adults. Applications such as GoHenry, FamZoo, Goalsetter, Step and FitMoney provide features related to budgeting, saving, spending, and in some cases parental monitoring.

Most of these applications require users to link real bank accounts or use prepaid debit cards. This allows users to manage actual money under supervision and gain practical exposure to financial management.

While these systems provide useful tools for money management, they are primarily designed for real-world financial transactions rather than structured financial education. Their interfaces and features are often oriented toward spending and banking activities, which may not be suitable for beginners who lack foundational financial knowledge.

3.1.1 Limitations of Existing Systems

The analysis of existing systems revealed several key limitations:

- Most applications require real money, exposing young users to potential financial risk.
- Financial education is often secondary to spending and banking functionalities.
- Many systems are region-specific and may not be fully accessible internationally.
- Limited use of structured learning paths or guided educational modules.
- Heavy reliance on parental supervision, which may reduce independent learning.
- Minimal focus on simulated financial practice in a risk-free environment.

These limitations indicate the need for a system that prioritizes structured financial education within a safe and controlled environment.

3.2 Comparative Analysis of Existing and Proposed System

A comparative analysis was conducted to evaluate existing financial applications against the proposed system. The comparison focuses on essential features related to financial learning, safety, and usability.

Feature	Existing Financial Applications	Proposed System
Target Audience	Children/teens with parental supervision	Youth and beginners
Use of Real Money	Requires real funds or bank-linked accounts	Uses simulated financial data only
Financial Risk	Risk of overspending or misuse	No financial risk

Educational Focus	Secondary to banking features	Primary focus on structured learning
Gamification	Limited reward-based elements	Structured gamification and progress tracking
Accessibility	Often region-specific	Designed for broader accessibility
Learning Environment	Real-world money management	Safe, controlled learning environment

The comparative analysis demonstrates that although existing systems provide practical financial tools, they do not sufficiently support beginners who require a structured and risk-free learning approach. The proposed system addresses these shortcomings by emphasizing education, engagement, and safety.

3.3 Description of the Proposed System

The proposed system is a mobile-based financial literacy application designed to educate young users on essential financial concepts such as budgeting, saving, goal setting, and responsible spending.

Unlike existing systems, the application operates using simulated financial data. This allows users to practice financial decision-making without the risks associated with real money.

The system includes:

- Interactive educational lessons.
- Virtual budgeting and expense tracking.
- Financial goal setting features.
- Gamified challenges and achievement tracking.
- Progress monitoring dashboard.

The application is designed to be intuitive, beginner-friendly, and accessible, encouraging independent learning while maintaining data privacy and security.

3.4 Functional Requirements

The functional requirements define the core features of the system:

FR 1: The system shall allow users to create an account and sign in using email and password.
FR 1.1: The system shall allow users to request a password reset via email.

FR 2: The system shall offer educational modules and lessons on budgeting, saving, debt management, and financial planning.
FR 2.1: The system shall include interactive learning elements such as quizzes and examples within the modules.

FR 3: The system shall allow users to track personal expenses, daily budgets, and savings goals.
FR 3.1: The system shall display spending summaries and budget progress.
FR 3.2: The system shall allow users to set savings goals and track progress over time.

FR 4: The system shall provide digital financial tools, including a virtual wallet simulation and financial calculators.
FR 4.1: The system shall allow users to simulate financial transactions within the wallet, such as adding funds, sending money, paying bills, and scanning QR codes for payments.
FR 4.2: The system shall provide financial calculators including a savings calculator, loan calculator, currency converter, and bill splitter.

FR 5: The system shall provide an AI assistant to answer financial literacy questions and assist users with learning topics.

FR 6: The system shall provide gamified learning elements including XP rewards, streak tracking, and progress tracking for completed modules.

FR 7: The system shall provide a summary dashboard displaying learning progress, streaks, XP rewards, financial insights, and quick access to tools.

FR 8: The system shall allow users to update their profile information.
FR 8.1: The system shall allow users to change their password securely.
FR 8.2: The system shall allow users to log out of the application.

FR 9: The system shall provide a rewards feature where users can earn XP and access deals or coupons based on their progress.

FR 10: The system shall provide onboarding guidance for first-time users to introduce the main features and functionalities of the application.

FR 11: The system shall provide notifications and financial tips to encourage users to develop better financial habits.

3.5 Non-Functional Requirements

The non-functional requirements describe the quality attributes of the system:

NFR 1: The system shall load the main interface within 2–3 seconds under normal network conditions.

NFR 2: The system shall be available on both iOS and Android platforms.

NFR 3: The system shall provide an intuitive and easy-to-navigate user interface.

NFR 4: The system shall ensure that all user authentication and financial data are protected using secure encryption mechanisms.

NFR 5: The system shall maintain visual consistency through a user-friendly color scheme and clear typography.

NFR 6: The system shall handle unstable internet connections gracefully and synchronize user data when connectivity is restored.

NFR 7: The system shall support real-time updates for notifications and user progress.

NFR 8: The system shall maintain high reliability with minimal downtime.

NFR 9: The system shall ensure secure storage of all personal and financial data in the backend database.

3.6 System Users

The proposed system will involve the following primary user:

- **Learners (Youth Users):** Individuals who use the application to learn financial concepts through interactive lessons, simulated budgeting activities, and goal-setting features.

Learners will be able to register, access educational modules, create virtual budgets, set financial goals, and track their progress within the application. Access to system functionalities will be restricted to authenticated users to ensure secure and efficient operation.

3.7 System Constraints and Assumptions

The system operates under the following constraints and assumptions:

- Users must have access to a smartphone and internet connection.
- Users are assumed to possess basic literacy and smartphone navigation skills.
- The system may rely on third-party services for authentication and backend services.

- The application is limited to simulated financial data and does not connect to real banking institutions.

3.8 Comparative Analysis of Development Approaches

Mobile applications can be developed using either native or cross-platform approaches. A comparison was conducted to determine the most suitable development strategy.

Feature	Native Development	Cross-Platform Development
Code Base	Separate code base per platform	Single shared code base
Development Time	Longer	Shorter
Development Cost	Higher	Lower
Performance	Very High	High
Maintenance	Complex	Easier

Cross-platform development approach is chosen for this project.
The application shall be available to users regardless of the type of smartphone they use, whether Android or iOS. Since a single shared code base will be used for both platforms, development time and cost will be significantly reduced. Additionally, maintaining and updating one code base simplifies future enhancements and bug fixes. Therefore, the cross-platform approach provides an efficient, cost-effective, and sustainable solution for the implementation of the proposed system.

3.9 Comparative Analysis of Mobile Application Development Frameworks

Several cross-platform frameworks were analysed, including Flutter, React Native, and Xamarin.

Feature	Flutter	React Native	Xamarin
Programming Language	Dart	JavaScript	C#
Performance	High	Medium	Medium
UI Design	Rich built-in widgets	Native components	Native components
Hot Reload	Yes	Yes	Yes
Community Support	Strong	Strong	Moderate

Flutter is chosen as the development framework for this project.
Its rich set of built-in widgets and high performance allow the creation of an intuitive and responsive user interface. The strong community support and single code base for both Android and iOS platforms further reduce development time and simplify maintenance. Additionally, the hot reload feature accelerates testing and debugging, ensuring efficient implementation of new features. Therefore, Flutter provides a reliable, flexible, and efficient framework for the development of the proposed financial literacy application.

3.10 Comparative Analysis of Database Systems

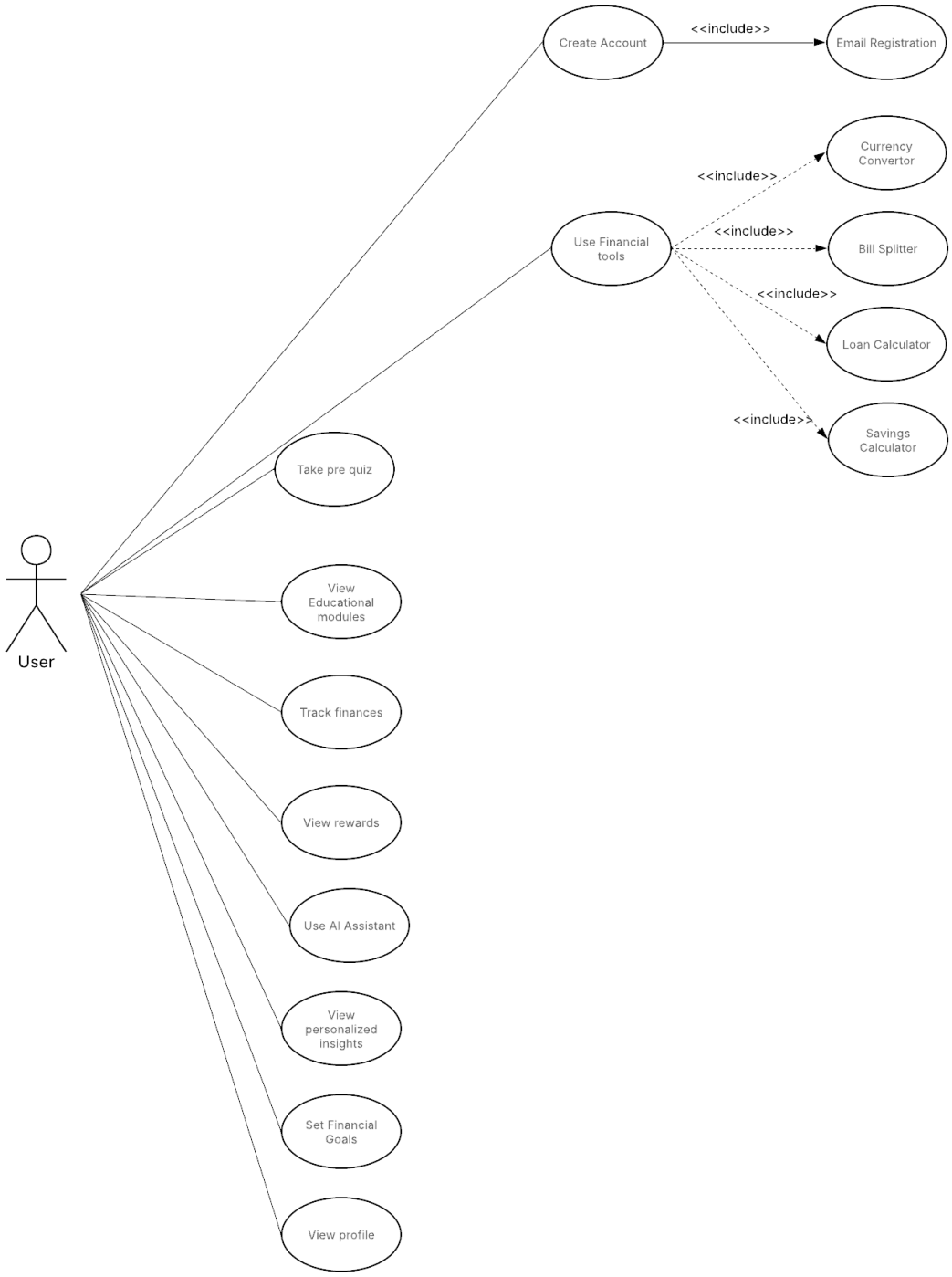
A database system is required to store user data, progress records, and simulated financial information. Three options were analysed: SQLite, Firebase, and PostgreSQL.

Feature	SQLite	Firebase	PostgreSQL
Database Type	Relational	NoSQL	Relational
Cloud Hosted	No	Yes	Yes
Real-Time Sync	No	Yes	No
Scalability	Low	High	High
Security	Medium	High	High

Firebase is selected as the database system for this project.
Its real-time data synchronization feature allows immediate updates of user progress and interactive activities, enhancing the responsiveness of the application. Firebase also offers

strong security measures and scalability, ensuring that user data is protected and that the system can handle an increasing number of users. The cloud-hosted nature of Firebase eliminates the need for complex server management, making it a practical and efficient solution for the proposed system's backend requirements.

3.11 Use Case Diagram



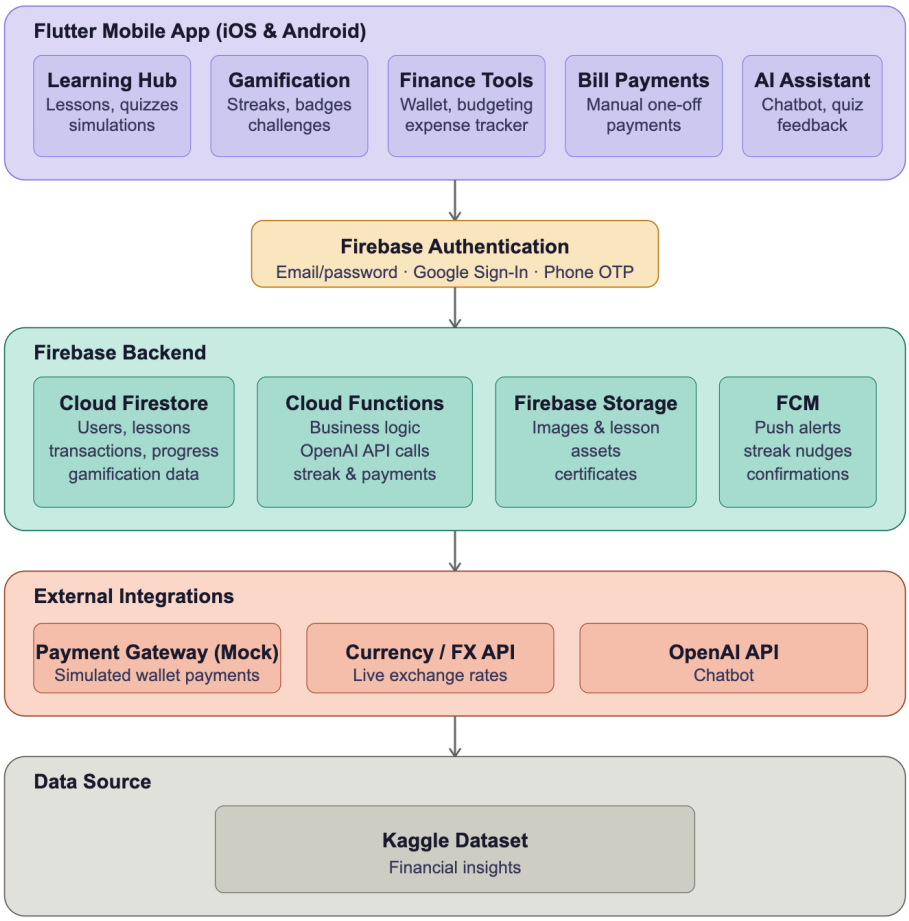
Chapter 4: Analysis

4.0 Introduction

This chapter presents the system design of the proposed financial literacy mobile application. System design translates the requirements identified in the analysis phase into a structured blueprint for implementation. It defines the architecture, components, and interactions within the system.

Several design techniques are used to represent the system structure and functionality, including the system architecture diagram, sitemap, data flow diagrams (DFD), use case diagram, database design, entity relationship diagram (ERD), flowcharts, and interface design. These models provide a clear overview of how the application operates and how different components interact to support the system functionalities.

4.1 System Architecture



1. Architecture Overview

The proposed financial literacy mobile application follows a layered architecture consisting of a mobile application layer, a backend services layer, an external services layer, and a data source layer.

- **Frontend (Flutter Mobile Application):** The user interface where learners interact with application features such as financial lessons, budgeting tools, goals, an AI-powered chatbot, and gamified learning elements.
- **Backend (Firebase Services):** Handles user authentication, data storage, progress tracking, server-side business logic, and notification services. Firebase ensures secure data management and real-time updates.
- **External Services:** Integrations including a currency exchange API for financial simulations, a mock payment gateway for simulated wallet transactions, and the OpenAI API which powers the in-app AI chatbot assistant.
- **Data Source (Kaggle Dataset):** A pre-loaded financial dataset sourced from Kaggle that provides financial insights used to enrich learning content and inform AI-assisted responses within the application.

This architecture ensures scalability, security, and efficient interaction between all application components.

2. Key Components

A. Mobile Application (Flutter Frontend)

The Flutter mobile application provides the interface through which users interact with the system. A single Dart codebase is compiled natively to both iOS and Android platforms.

- **Learning Hub**

  Provides financial literacy lessons, quizzes, and interactive learning scenarios to help users understand financial concepts.

- **Gamification Module**

  Includes features such as streaks, badges, and experience points (XP) to motivate users and increase engagement with learning activities.

- **Finance Tools**

  Offers budgeting tools, wallet simulation, and expense tracking features that allow users to practise financial management in a safe environment.

- **Bill Payments**

  Allows users to simulate manual one-off bill payments within the app using the mock payment gateway, reinforcing real-world financial habits without using real money.

- **AI Assistant**

An embedded chatbot powered by the OpenAI API that answers financial literacy questions, provides quiz feedback, and assists users in understanding financial concepts in a conversational format.

- **User Profile**

  Allows users to manage their account information, view progress, and adjust application settings.

B. Firebase Backend Services

Firebase provides the backend infrastructure required to support the mobile application. All communication between Flutter and Firebase occurs via the FlutterFire SDK.

- **Firebase Authentication**

  Manages user registration and login using secure authentication methods including email/password and Google Sign-In.

- **Cloud Firestore**

  Stores all user data including profiles, financial goals, lesson progress, expense records, and gamification achievements. Flutter widgets subscribe to Firestore streams for real-time UI updates.

- **Cloud Functions**

  Handles all server-side logic including updating streaks, processing rewards, managing automated tasks, processing mock payments, and securely forwarding requests to the OpenAI API so that API keys are never exposed on the client.

- **Firebase Storage**

  Stores application assets such as lesson images, completion certificates, and educational content files.

- **Firebase Cloud Messaging (FCM)**

  Sends push notifications to users including streak reminders, payment confirmations, badge unlock alerts, and engagement nudges.

C. External Integrations

External integrations provide additional functionality to enhance the application experience.

- **Payment Gateway (Mock / Simulated)**

  A simulated payment engine that processes wallet payments within the app. This allows full end-to-end testing of payment flows without a live gateway integration, making it appropriate for a prototype application.

- **Currency / Foreign Exchange API**

Provides real-time exchange rate information used within financial tools such as currency converters and financial simulations, improving the realism of the learning environment.

- **OpenAI API**

  Powers the in-app AI chatbot assistant. Requests are routed through Cloud Functions to keep API credentials secure on the server side. The chatbot answers financial literacy questions, provides contextual quiz feedback, and supports learners throughout their financial education journey.

D. Data Source — Kaggle Dataset

A financial dataset sourced from Kaggle is pre-loaded into the system to provide financial insights. This dataset informs the AI assistant's responses and enriches the financial learning content presented to users. By grounding the chatbot and lesson recommendations in real financial data, the application provides more meaningful and context-aware guidance to learners.

3. Workflow Example

The following example illustrates how the system processes a typical user interaction from login through to feature use.

1. User Accesses the Application

The user opens the mobile application and logs in using their credentials or Google Sign-In.

2. Authentication Process

The Flutter application sends the login request to Firebase Authentication. Firebase verifies the user credentials and grants access. The user's profile document is fetched from Cloud Firestore and the home dashboard is loaded.

3. User Interaction with Features

The user may access lessons, interact with the AI chatbot, track expenses, make simulated bill payments, or create financial goals. The application retrieves and updates relevant data through Cloud Firestore in real time.

4. AI Chatbot Interaction

When the user submits a question to the AI Assistant, the Flutter app calls a Cloud Function. The Cloud Function securely forwards the query along with relevant financial context (drawn from the Kaggle dataset and the user's progress) to the OpenAI API. The response is returned to the user through the chatbot interface.

5. Data Processing

Cloud Functions process backend actions such as updating XP rewards, calculating streak progress, recording simulated payment transactions, and maintaining learning streaks.

6. Notifications

Firebase Cloud Messaging sends reminders or alerts to encourage continued engagement, including streak risk warnings, payment confirmations, and badge unlock notifications.

4. Security Considerations

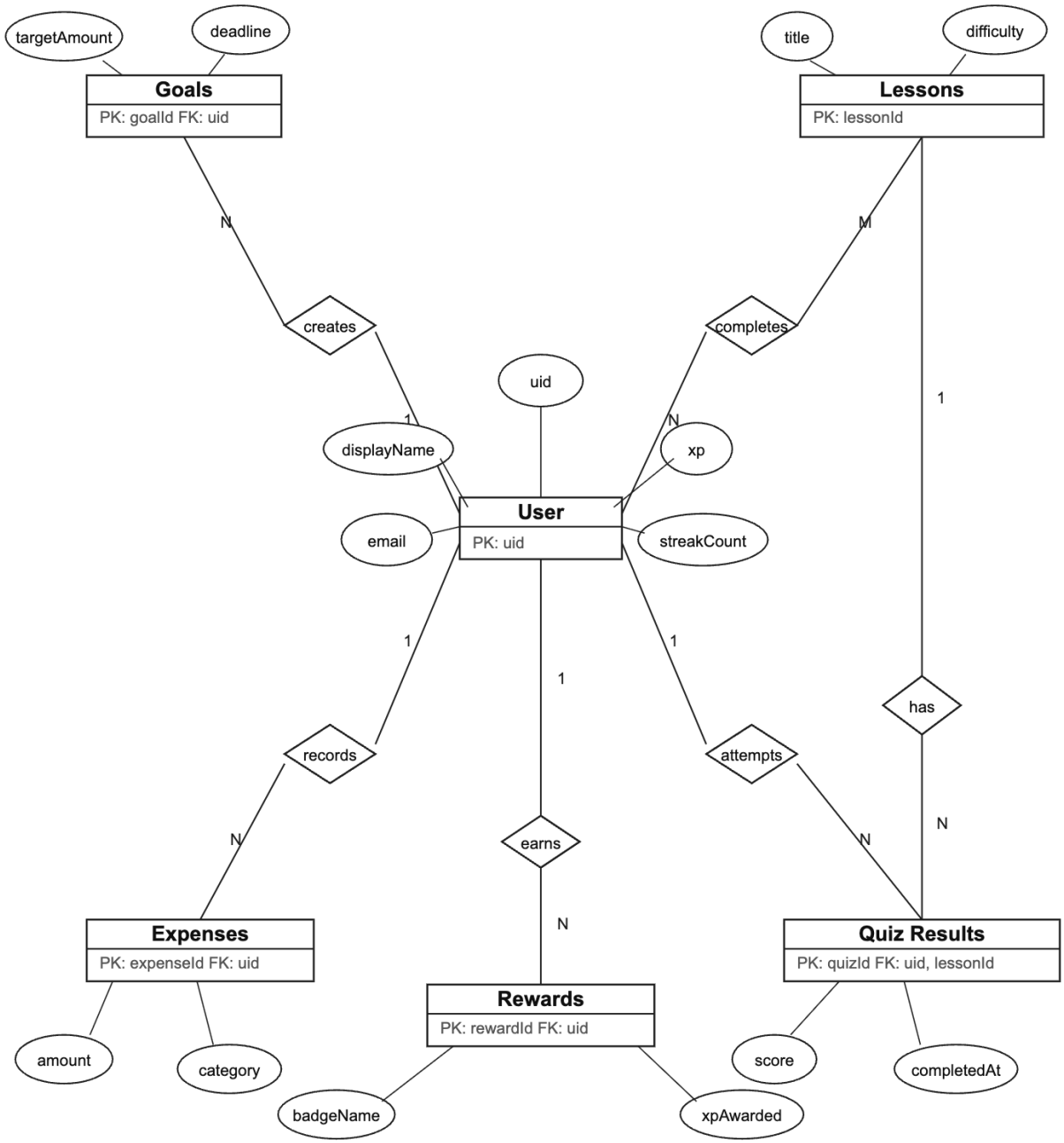
Several security measures are implemented to protect user data and ensure safe system operation.

- **Secure Authentication:** Firebase Authentication ensures that only authorised users can access the application, using industry-standard OAuth and JWT-based token management.
- **Encrypted Communication:** Data exchanged between the mobile application and backend services is protected using secure HTTPS protocols enforced by Firebase.
- **Secure Data Storage:** Cloud Firestore enforces security rules based on the authenticated user's UID, preventing unauthorised access to or modification of user data.
- **API Key Protection:** The OpenAI API key is stored exclusively in Cloud Functions environment configuration and is never embedded in the Flutter client application.
- **Mock Payment Safety:** All payment flows use a simulated gateway with no real financial transactions, eliminating any financial risk for users during the prototype phase.

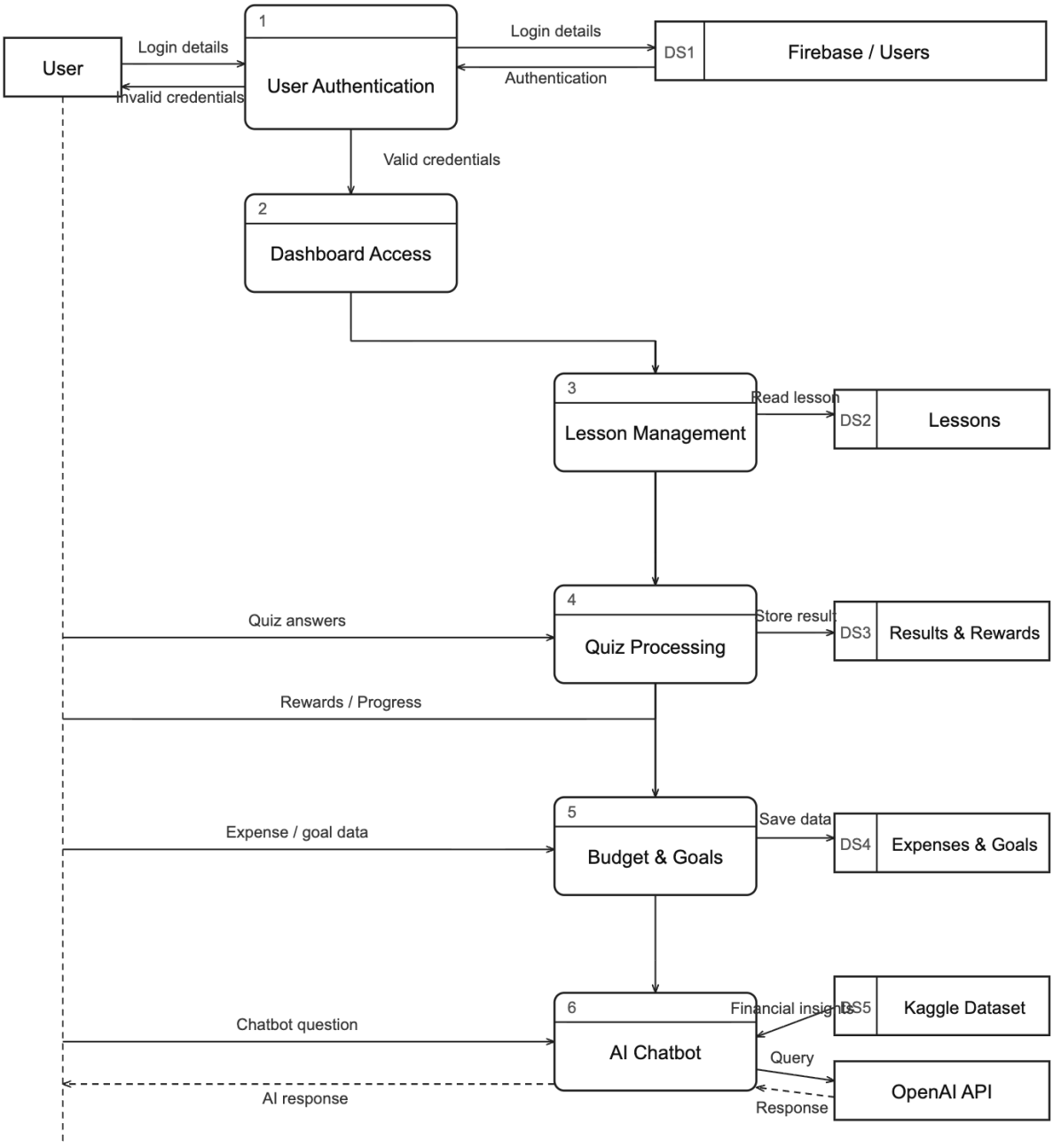
Conclusion

The proposed architecture provides a scalable and secure structure for the financial literacy mobile application. By combining a Flutter-based frontend with Firebase backend services, an AI-powered chatbot via the OpenAI API, and a Kaggle-sourced financial dataset, the system supports efficient data management, real-time updates, and engaging user features including gamification, financial simulations, and personalised AI assistance. This architecture ensures that users can interact with the application seamlessly while maintaining secure, reliable, and educationally meaningful system operations.

4.2 Entity Relationship Diagram (ERD)

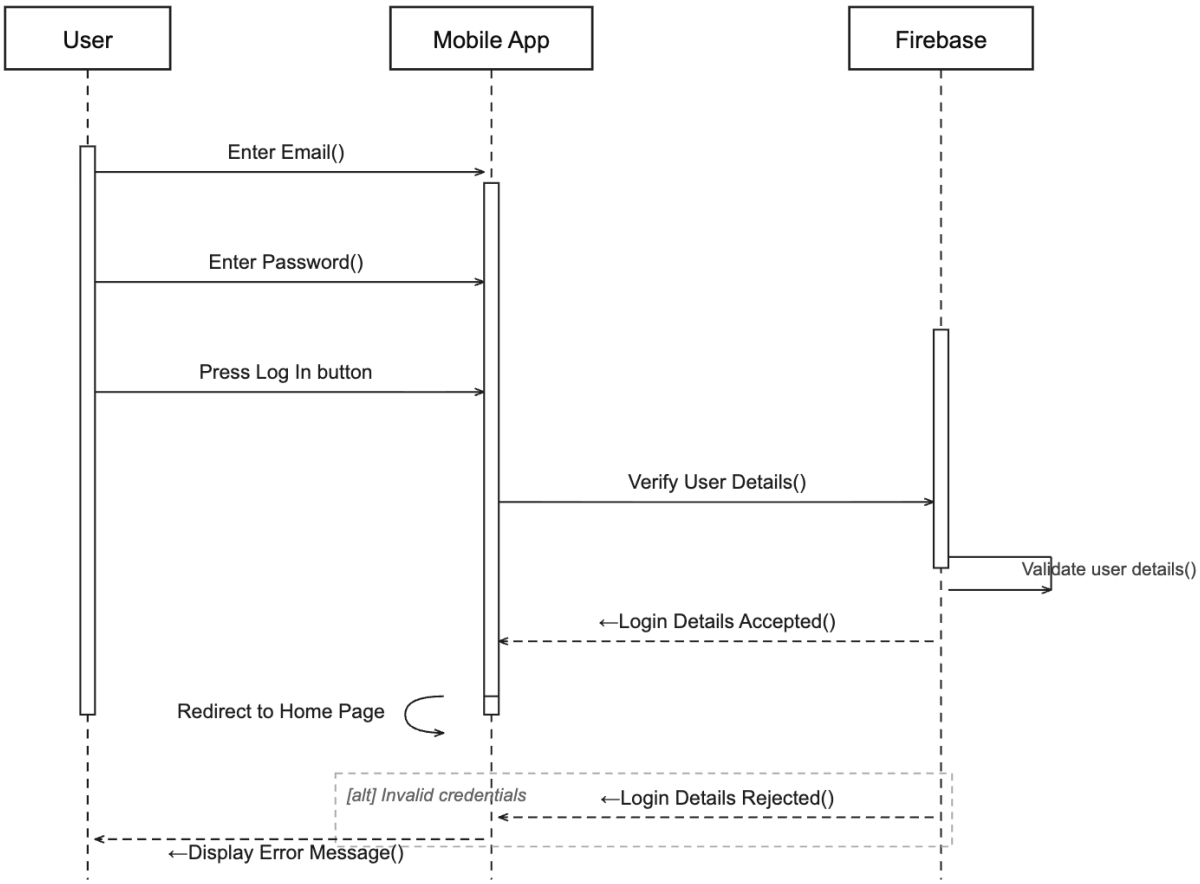

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score
completedAt

4.3 Data Flow Diagram (DFD)

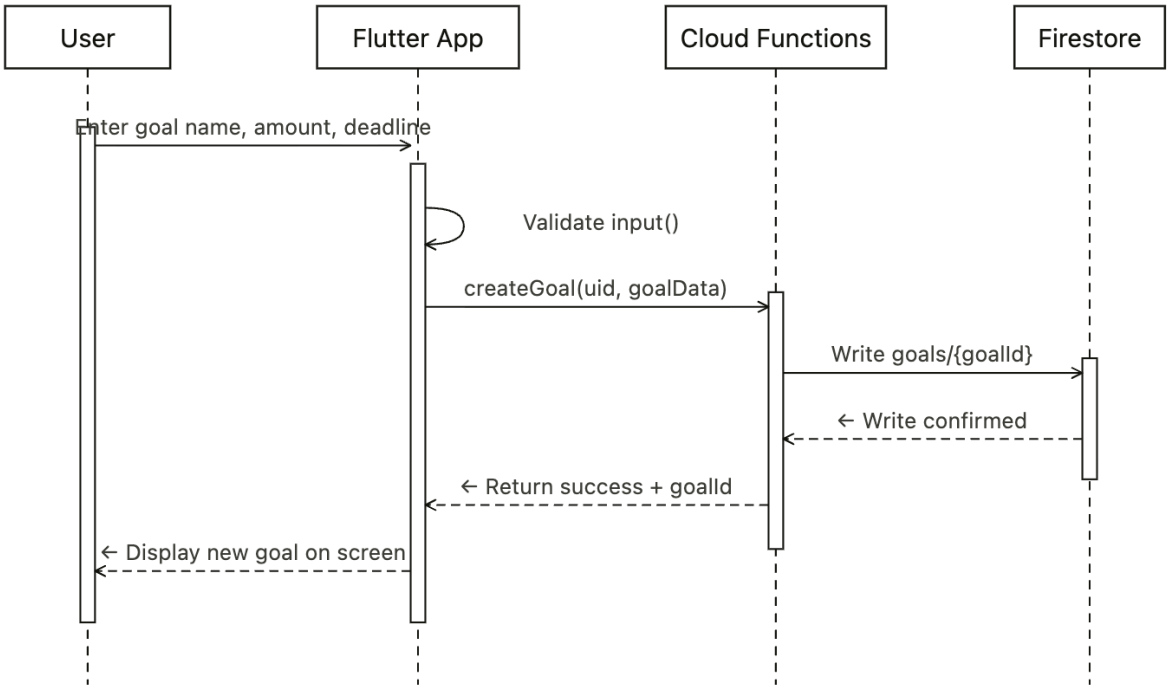


4.4 Sequence Diagram

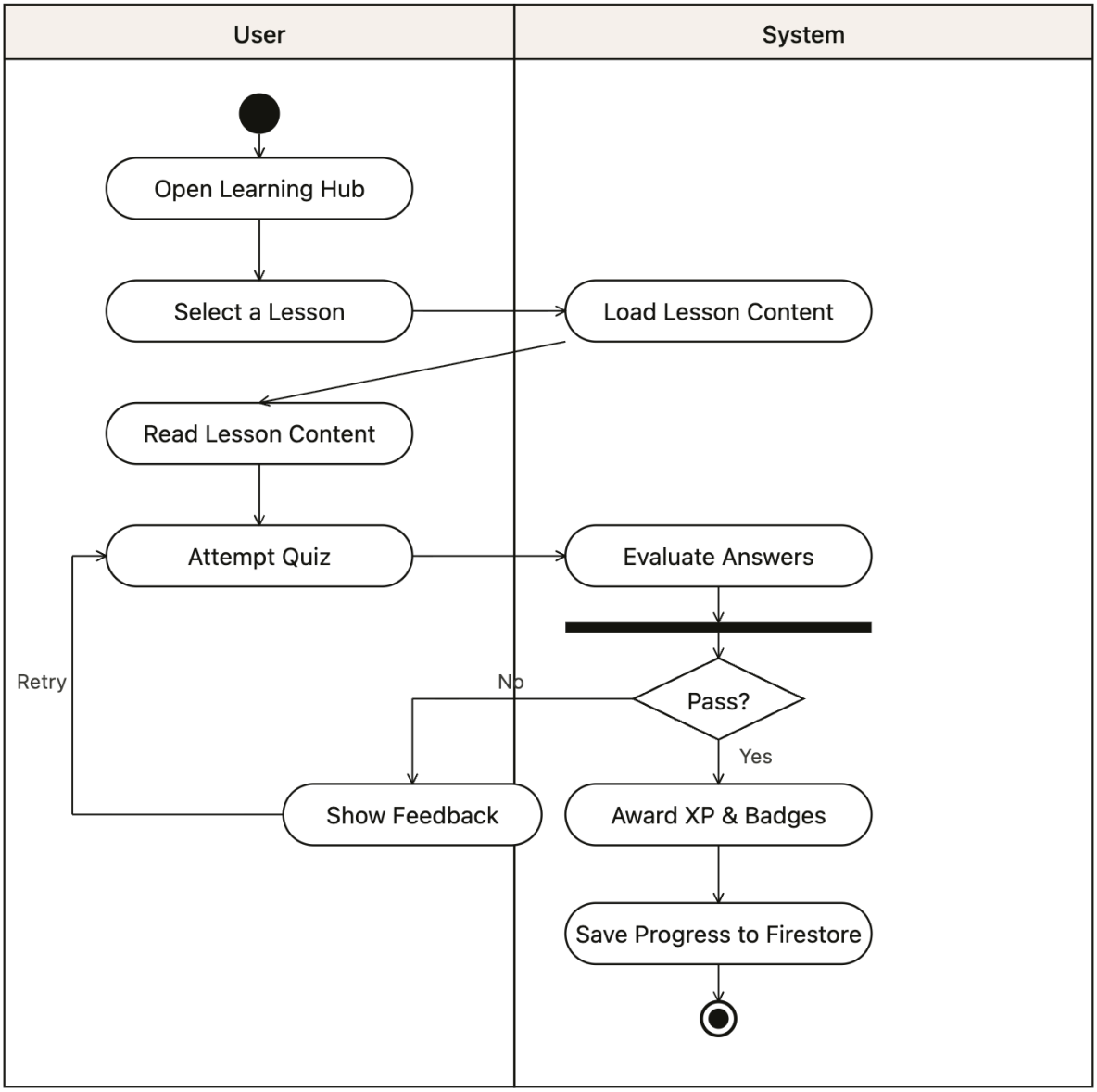
4.4.1 Login Sequence Diagram



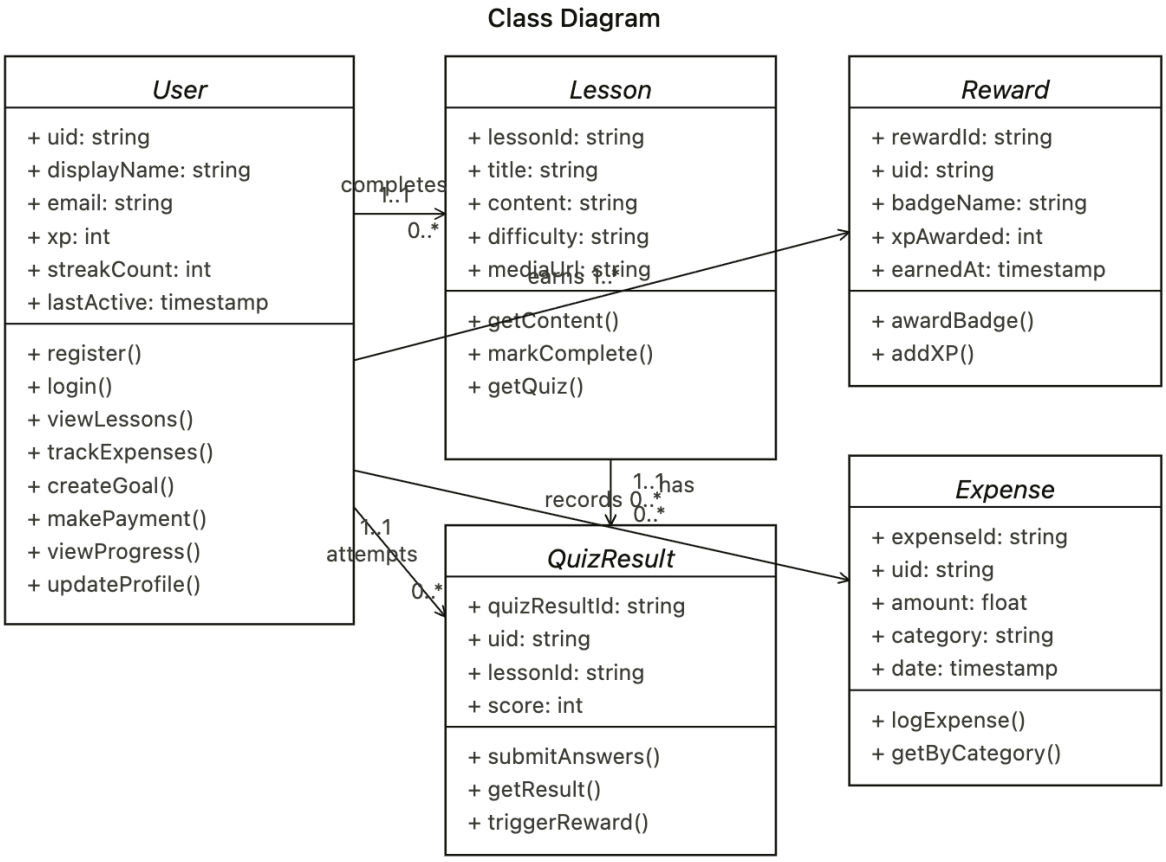
4.4.1 Goal Creation Sequence Diagram


User
Flutter App
Cloud Functions
Firestore
Enter goal name, amount, deadline
Validate input()
createGoal(uid, goalData)
Write goals/{goalId}
← Write confirmed
← Return success + goalId
← Display new goal on screen

4.5 Activity Diagram



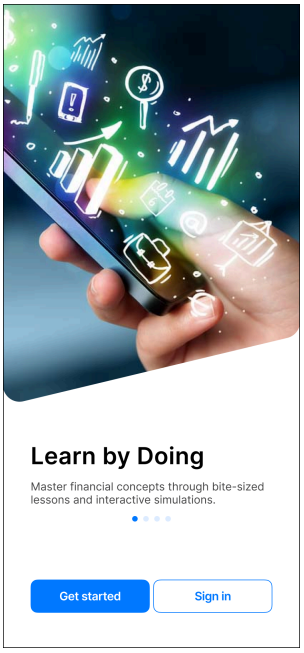
4.6 Class Diagram



4.7 UI/UX Design

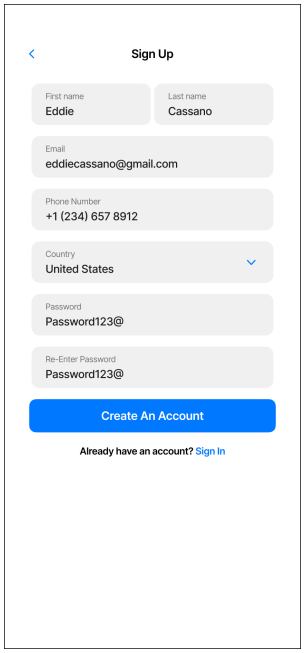
4.7.1 Landing page

Purpose: First screen introducing the application.



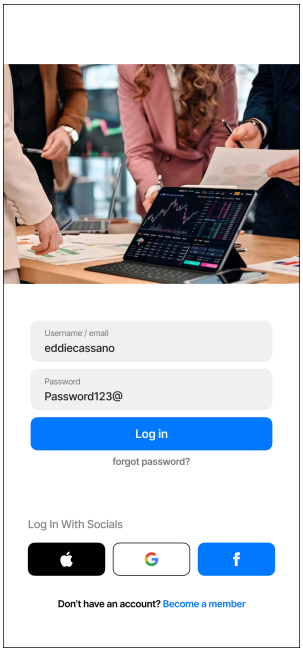
4.7.2 Register page

Purpose: Allows new users to create an account.



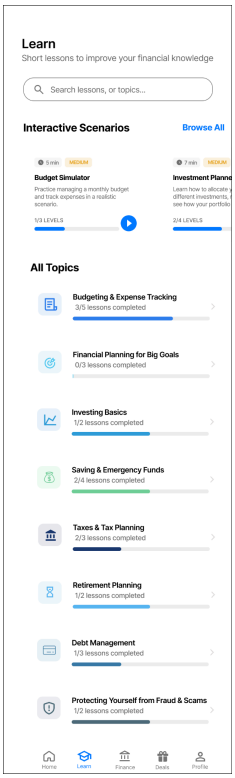
4.7.3 Login page

Purpose: Allows existing users to sign into the system.



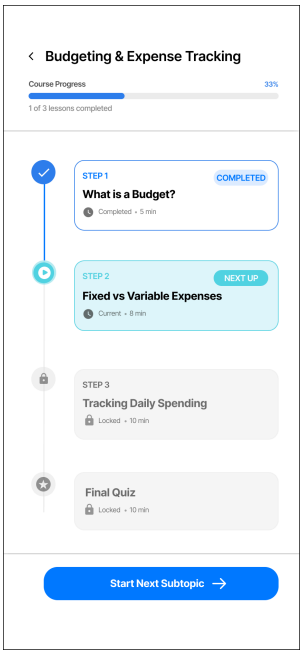
4.7.4 Learn page

Purpose: Allows existing users to sign into the system.



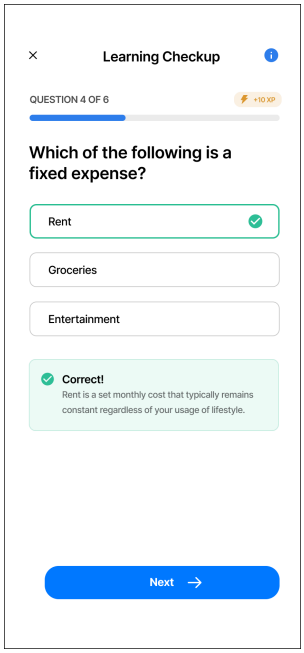
4.7.5 Subtopic page

Purpose: Displays detailed financial learning content.



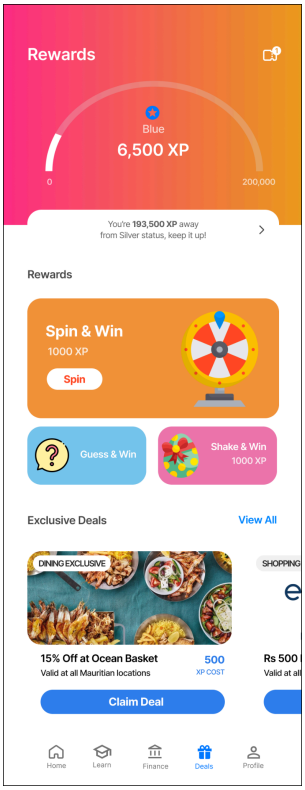
4.7.6 Quiz page

Purpose: Displays detailed financial learning content.



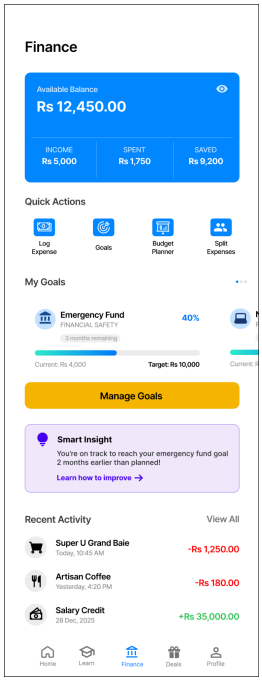
4.7.7 Rewards page

Purpose: Displays gamification elements such as XP points



4.7.8 Finance page

Purpose: Displays finance tools



Chapter 5: Implementation

5.0 Introduction

This chapter describes the implementation of the proposed financial literacy mobile application. It presents the hardware and software tools used during development and explains how the different components of the system were implemented. The chapter also outlines the programming standards followed during development and describes the implementation of the major modules of the application.

5.1 Development Tools and Environment Used

5.1.1 Hardware

The specifications of the laptop used to develop and compile the application are as follows:

1. Processor – Apple M2
2. RAM – 8 GB
3. Storage – 256 GB SSD

The specifications of the mobile device used to test the application are as follows:

1. Operating System – iOS
2. Processor – Apple A13 Bionic chip
3. Sensors – Camera
4. Connectivity – WiFi

5.1.2 Software and Languages

To develop the financial literacy mobile application, several software tools and programming technologies were used.

1. Flutter SDK

Flutter is a cross-platform development framework used to build mobile applications for both Android and iOS using a single codebase. It provides a rich set of widgets that allow developers to design responsive and modern user interfaces efficiently.

2. Dart Programming Language

Dart is the programming language used in Flutter development. It supports object-oriented programming and enables developers to build high-performance mobile applications.

3. Firebase

Firebase was used as the backend service for the application. It provides authentication services and a cloud database used to store user information, financial goals, lessons, and quiz results.

4. Visual Studio Code

Visual Studio Code was used as the primary development environment for writing and managing the Flutter application code. It provides useful extensions for Flutter and Dart development.

5. Android Studio

Android Studio was used to run Android emulators and test the application on Android devices during development.

6. Xcode

Xcode was used for building and testing the application on iOS devices. It provides the required tools for compiling Flutter applications for iOS and running the iOS simulator.

5.2 Programming Standards

To maintain code readability and consistency, certain programming standards were followed during the development of the application.

5.2.1 Naming conventions

The naming conventions were applied to ensure that the source code remains organized and easy to understand for future developers.

Type of Identifier	Naming Rule	Example
Classes	Start with an uppercase letter	HomeScreen, LoginScreen
Methods	Start with lowercase letter and represent actions	loginUser(), createGoal()
Variables	Start with lowercase letter and be meaningful	userName, goalAmount

5.3 Implementation of Modules

This section explains how the main modules of the financial literacy application were implemented.

5.3.1 Onboarding Page

The onboarding page is the first interface displayed when users open the application. It introduces the financial literacy platform and provides options for users to either sign in to an existing account or create a new account. This screen serves as the entry point of the application and allows users to navigate to the authentication modules.

```
PageView.builder(
                  controller: _pageController,
                  onPageChanged: (index) {
                    setState(() {
                      _currentPage = index;
                    });
                  },
                  itemCount: pages.length,
                  itemBuilder: (context, index) {
                    return OnboardingContent(page: pages[index]);
                  },
                ),
```

5.3.2 User Registration Module

The user registration module allows new users to create an account within the application. Firebase Authentication was integrated to securely manage user account creation. Users are required to enter their personal information such as name, email, and password. Once the information is validated, the system registers the user and stores the account information securely.

```dart
Future<void> _handleSignUp() async {
    final firstName = _firstNameController.text.trim();
    final lastName = _lastNameController.text.trim();
    final email = _emailController.text.trim();
    final phone = _phoneController.text.trim();
    final password = _passwordController.text.trim();
    final rePassword = _reEnterPasswordController.text.trim();

    if (firstName.isEmpty ||
        lastName.isEmpty ||
        email.isEmpty ||
        phone.isEmpty ||
        _selectedCountry.isEmpty ||
        password.isEmpty ||
        rePassword.isEmpty) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Please fill in all fields.')),
      );
      return;
    }

    if (!_emailRegex.hasMatch(email)) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Please enter a valid email address.')),
      );
      return;
    }

    if (password.length < 6) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Password must be at least 6 characters.')),
      );
      return;
    }

    if (password != rePassword) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Passwords do not match.')),
      );
      return;
    }

    setState(() {
      _isLoading = true;
    });

  final user = await _authService.registerUser(
  firstName,
  lastName,
  email,
  phone,
  _selectedCountry,
  password,
);

    if (!mounted) return;

    setState(() {
      _isLoading = false;
    });

    if (user != null) {
        Navigator.of(context).pushReplacement(
          MaterialPageRoute(builder: (context) => PreQuizScreen()),
        );
        return;
    }
```

5.3.3 User Login Module

The login module allows registered users to access the application using their credentials. Firebase Authentication verifies the user's email and password before granting access to the system. If the credentials are correct, the user is redirected to the home screen of the application.

```dart
Future<void> _handleLogin() async {
    final email = _usernameController.text.trim();
    final password = _passwordController.text.trim();

    if (email.isEmpty || password.isEmpty) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Please enter email and password.')),
      );
      return;
    }

    if (!_emailRegex.hasMatch(email)) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text('Please enter a valid email address.')),
      );
      return;
    }

    setState(() {
      _isLoading = true;
    });

    final user = await _authService.loginUser(email, password);

    if (!mounted) return;

    setState(() {
      _isLoading = false;
    });

    if (user != null) {
      Navigator.of(context).pushReplacement(
        MaterialPageRoute(builder: (context) => const HomeScreen()),
      );
      return;
    }
```

5.3.4 Learning Module

The learning module provides educational content related to financial literacy. Lessons are organised into topics and subtopics to help users understand financial concepts in a structured way. Users can browse through different lessons and select the topic they wish to learn.



```
onTap: () async {
            var progressData =
                await TopicProgressService.instance.getTopicProgress(topicId);
            var preQuizDone = progressData?['preQuizCompleted'] == true;
            var shouldAutoStartFlow = false;

            if (!context.mounted) return;

            if (!preQuizDone) {
              await Navigator.push(
                context,
                MaterialPageRoute(
                  builder: (context) => QuizScreen(
                    quizTitle: 'Before You Start Quiz',
                    questions: _getPreQuizQuestionsForTopic(title),
                    awardXp: false,
                    showCompletionFlow: false,
                    revealAnswerFeedback: false,
                    onQuizFinished: (score, totalQuestions) {
                      return TopicProgressService.instance
                          .savePreQuizResult(
                            topicId: topicId,
                            score: score,
                            totalQuestions: totalQuestions,
                          )
                          .then((_) async {
                        await XpService.instance.earnXp(
                          amount: 10,
                          source: 'pre_quiz_completion',
                        );
                      });
                    },
                  ),
                ),
              );

              progressData =
                  await TopicProgressService.instance.getTopicProgress(topicId);
              preQuizDone = progressData?['preQuizCompleted'] == true;

              if (!preQuizDone) return;
              shouldAutoStartFlow = true;
            }

            if (!context.mounted) return;

            final persistedLessonsCompleted =
                (progressData?['lessonsCompleted'] as int?)?.clamp(
                      0,
                      totalLessons,
                    ) ??
                    0;

            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) => SubtopicDetailScreen(
                  topicId: topicId,
                  preQuizCompleted: preQuizDone,
                  autoStartFlow: shouldAutoStartFlow,
                  topicTitle: title,
                  currentLesson: persistedLessonsCompleted,
                  totalLessons: totalLessons,
                  lessons: _getLessonsForTopic(title),
                ),
              ),
            );
          },
```

The subtopic module presents the detailed content of each financial learning topic. Before accessing the lesson content, users are required to complete a short pre-quiz, which evaluates their initial understanding of the topic. After completing the pre-quiz, users can progress through

structured lesson steps that explain financial concepts such as budgeting, saving, and financial planning. Once all lessons are completed, users take a final quiz to assess their knowledge and reinforce learning.

```
onTap: lessonStatus != LessonStatus.locked
                    ? () {
                        if (isPreQuiz && _preQuizCompleted) {
                          return;
                        }

                        // Check if it's a quiz
                        if (lessonStatus == LessonStatus.quiz || isPreQuiz || isFinalQuiz) {
                          Navigator.push(
                            context,
                            MaterialPageRoute(
                              builder: (context) => QuizScreen(
                                quizTitle: isPreQuiz ? 'Before You Start Quiz' : 'Learning Checkup',
                                questions: isPreQuiz
                                  ? _getPreQuizQuestions()
                                  : _getPostQuizQuestions(),
                                awardXp: !isPreQuiz,
                                showCompletionFlow: !isPreQuiz,
                                revealAnswerFeedback: !isPreQuiz,
                                topicId: widget.topicId,
                                topicTitle: widget.topicTitle,
                                lessonsCompleted: _coreLessonCount,
                                totalLessons: _coreLessonCount,
                                onQuizFinished: (score, totalQuestions) async {
                                  if (isPreQuiz) {
                                    return TopicProgressService.instance.savePreQuizResult(
                                      topicId: widget.topicId,
                                      score: score,
                                      totalQuestions: totalQuestions,
                                    );
                                  } else {
                                    await TopicProgressService.instance.savePostQuizResult(
                                      topicId: widget.topicId,
                                      score: score,
                                      totalQuestions: totalQuestions,
                                    );
                                    // Award XP for quiz completion (not pre-quiz)
                                    await XpService.instance.earnXp(amount: 50, source: 'quiz_completion');
                                  }
                                },
                              ),
                            ),
                          )
```

5.3.6 Quiz Module

The quiz module evaluates the user's understanding of financial concepts presented in the lessons. Users answer multiple-choice questions, and the system checks whether the selected answer is correct. For correct answers, XP points are awarded and the score is updated. At the end of the quiz, the system processes the results and navigates the user to the quiz completion screen, where performance and rewards are displayed.

```dart
void _submitAnswer({required bool markAnswered}) {
    if (_selectedAnswer == _currentQuestion.correctAnswer) {
      _score++;
      if (widget.awardXp) {
        _totalXP += 10;
        _showCorrectXpPopup();
        _persistCorrectAnswerXp();
      }
    }

    if (markAnswered) {
      setState(() {
        _answered = true;
      });
    }
```

```dart
Future<void> _showResults() async {
    if (widget.onQuizFinished != null) {
      try {
        await widget.onQuizFinished!(_score, widget.questions.length);
      } catch (error) {
        debugPrint('Failed running quiz finish callback: $error');
      }
    }

    if (!widget.showCompletionFlow) {
      if (!mounted) {
        return;
      }

      Navigator.of(context).pop({
        'score': _score,
        'totalQuestions': widget.questions.length,
      });
      return;
    }

    var currentLevel = 1;
    var xpToNextLevel = _xpPerLevel;
    int? preQuizScore;
    int? preQuizTotal;

    try {
      final totalXp = await XpService.instance.getXp();
      final xpInCurrentLevel = totalXp % _xpPerLevel;
      currentLevel = (totalXp ~/ _xpPerLevel) + 1;
      xpToNextLevel = xpInCurrentLevel == 0
          ? _xpPerLevel
          : _xpPerLevel - xpInCurrentLevel;
    } catch (error) {
      debugPrint('Failed to load XP progress for completion screen: $error');
    }

    if (widget.topicId != null) {
      try {
        final progress = await TopicProgressService.instance.getTopicProgress(
          widget.topicId!,
        );
        final rawPreScore = progress?['preQuizScore'];
        final rawPreTotal = progress?['preQuizTotal'];

        if (rawPreScore is int && rawPreTotal is int && rawPreTotal > 0) {
          preQuizScore = rawPreScore;
          preQuizTotal = rawPreTotal;
        }
      } catch (error) {
        debugPrint('Failed to load pre-quiz progress for completion screen: $error');
      }
    }

    if (!mounted) {
      return;
    }

    // Navigate to completion screen
    Navigator.of(context).pushReplacement(
      MaterialPageRoute(
        builder: (context) => QuizCompletionScreen(
          score: _score,
          totalQuestions: widget.questions.length,
          earnedXP: _totalXP,
          currentLevel: currentLevel,
          xpToNextLevel: xpToNextLevel,
          topicTitle: widget.topicTitle,
          lessonsCompleted: widget.lessonsCompleted,
          totalLessons: widget.totalLessons,
          preQuizScore: preQuizScore,
          preQuizTotal: preQuizTotal,
        ),
      ),
    );
  }
```

5.3.7 Rewards Module

The rewards module displays the user's accumulated experience points (XP) and unlockable reward features within the application. The screen retrieves XP data in real time through the XpService, allowing the interface to update dynamically as the user completes quizzes and lessons. In addition, the module presents reward-based activities and exclusive deals, encouraging continued engagement with the application.

```dart
return StreamBuilder<int>(
      stream: XpService.instance.xpStream(),
      builder: (context, snapshot) {
        final currentXp = snapshot.data ?? 0;

        return Scaffold(
          body: Container(
            color: const Color(0xFFF0F0F0),
            child: SingleChildScrollView(
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  _buildGradientHeader(currentXp),
                  const SizedBox(height: 8),
                  _buildStatusCard(currentXp),
```

Chapter 6: Testing

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